

Do surface and bed observables determine basal friction?

A point-wise test in the Amundsen Sea sector

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ABSTRACT. Uncertainties in basal-sliding parameterizations hinder accurate projections of ice-sheet contributions to sea-level rise. Although inverse methods can calibrate basal friction to reproduce observed velocities, the controls on inversion-derived friction remain poorly understood. We test whether surface, bed, and geophysical observables can define a consistent point-wise relationship with the inversion-derived basal-friction control across the Amundsen Sea sector of West Antarctica. Using a sector-wide inversion for the basal-friction control as the reference, we train neural-network regression models in ten experiments, each with a spatially separated 50 km square and buffer withheld from training and validation. We test each friction-control field in a forward simulation against observed velocity. Depending on predictor configuration, the friction fields reduce modeled-velocity error relative to a spatially uniform friction reference in six to nine of the ten squares. Performance varies among locations, adding predictors does not consistently improve performance, and failures occur even where marginal and joint predictor support is high. Although the tested local predictors improve velocity in most withheld squares, modeled-velocity errors remain substantially larger than for the inversion reference. These results indicate that the considered surface, bed, and geophysical observables are insufficient to define a consistently successful point-wise parameterization of inversion-derived basal friction.

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INTRODUCTION

Ice-sheet models are used to project sea-level rise, reconstruct past ice-sheet behavior, and investigate the physical processes governing ice-mass change, e.g. Lecavalier and Tarasov (2025); Nowicki and others (2016). These models contain structural uncertainty arising from incomplete representations of physical processes and parameter uncertainty arising from imperfectly constrained model parameters (Hebeler and others, 2008; Recinos and others, 2023). Both sources of uncertainty limit our ability to constrain probabilistic projections, especially for low-probability, high-impact outcomes (Kopp and others, 2017).

Basal slip is an important source of both structural and parameter uncertainty (Joughin and others, 2019; Bulthuis and others, 2019). Motion at the ice–bed interface can involve sliding over hard bedrock, deformation of subglacial sediment, and changes in basal water pressure, among other processes (Schoof and Hewitt, 2013; Noble and others, 2020). Ice-sheet models generally represent these processes through a sliding law containing one or more spatially varying friction parameters (Weertman, 1964; Schoof, 2005). In an inversion, these parameters are adjusted to reduce the mismatch between modeled and observed quantities such as velocity (MacAyeal, 1993). The resulting friction field can therefore absorb unresolved basal processes, including bed roughness, till rheology, and hydrology (Tulaczyk and others, 2000; Lu and Kingslake, 2024), as well as geometry errors and differences among model formulations (Barnes and others, 2021). It is consequently an effective model parameter rather than a direct observation of basal conditions.

Inversion studies show that basal resistance across Antarctica is spatially heterogeneous (Morlighem and others, 2013; Zhao and others, 2018). This motivates a working hypothesis: at the resolution of an ice-sheet model, the dimensionless basal-friction control $C(\mathbf{x})$ may be approximated by a point-wise function of observables evaluated at the same location,

$$C(\mathbf{x}) \approx f(\mathbf{z}(\mathbf{x})), \quad (1)$$

where $\mathbf{z}(\mathbf{x})$ contains locally measured or inferred quantities that may influence basal resistance or serve as proxies for unresolved basal conditions. Equation 1 is a testable point-wise approximation at the model-grid scale, not a physical sliding law. We test it by fitting the relationship outside a held-out region and asking whether the predicted C improves modeled velocity within that region, particularly where similar predictor values are well represented in the training data.

We divide the candidate predictors into two groups. The ice-product group describes ice geometry,

its gradients, driving stress, and surface air temperature. The geophysical-product group describes bed geometry, bed–surface gradient alignment, geothermal heat flux, and gravity and magnetic anomalies. Ice thickness controls overburden pressure and normal stress at the bed, while surface slope governs driving stress; together, variations in thickness and surface elevation gradient impose systematic constraints on basal drag (MacAyeal, 1989; Cuffey and Paterson, 2010, Sections 7.2–7.4 and 8.2). Surface air temperature provides the upper thermal boundary condition and influences the ice sheet’s englacial and basal thermal state (Meierbachtol and others, 2015; Larour and others, 2012).

Geophysical factors can affect basal drag through bed topography and substrate composition. Bed gradients characterize topographic variation that can contribute to form drag, while deformable till can reduce resistance and enable sliding (Cuffey and Paterson, 2010; Tulaczyk and others, 2000; Boulton and others, 2001; Hoffman and others, 2022). Gravity disturbances reflect subsurface density variations associated with bedrock and sedimentary structure (Hackney and Featherstone, 2003). Gravity data can therefore constrain broad bed geometry where direct measurements are sparse (Boghosian and others, 2015; Kennedy, 2020). Bedrock lithology can influence sediment characteristics, while deformable sediment can reduce basal resistance (Crompton and Flowers, 2016; Boulton and others, 2001). Gravity measurements cannot resolve the thin (centimeter- to meter-scale) sediment layers that affect sliding, but larger density contrasts can identify sedimentary basins; such sediment bodies can influence ice-stream onset (Kennedy, 2020; Peters and others, 2006). Magnetic anomalies reflect contrasts in crustal magnetization (Golynsky and others, 2018). Together with gravity and radar data, they can help infer subglacial geology (Jordan and others, 2023) and regional thermal structure (Martos and others, 2017; Reading and others, 2022), providing indirect context for basal drag. Geothermal heat flux supplies heat at the bed and can influence whether basal ice reaches the pressure-melting point. Where basal ice is at the melting point, additional heat can support basal meltwater production and thereby influence basal resistance (Dawson and others, 2022; Kang and others, 2022).

Previous studies have compared inferred basal resistance with radar, seismic, and other observations of the subglacial environment, but have not established a generally predictive local relationship. Haris and others (2024) found differences in basal shear stress among regions with distinct radar signatures but did not identify a direct predictive relationship. Das and others (2023) found no robust correlation between inferred sliding parameters and radar-derived basal reflectivity over Thwaites Glacier, while Kyrke-Smith and others (2017) found only weak kilometer-scale association between basal drag and seismic acoustic

impedance along Pine Island Glacier (PIG).

These studies, together with our understanding of basal processes, suggest that the relationship between basal conditions and basal resistance may be complex and nonlinear, involving thermo-mechanical feedbacks, subglacial hydrology, and spatially variable bed properties. Following the field inversion and machine learning (FIML) approach of Parish and Duraisamy (2016), we use nonlinear regression with a multilayer perceptron (MLP) as a flexible representation of f in Eqn 1 to relate the selected observables to the inversion-derived friction control C .

The point-wise architecture asks whether the selected observables at a location contain enough information to determine the inversion control there, without prescribing a particular functional form. We deliberately exclude observed velocity from the predictor set because changing C changes the modeled velocity; including velocity as an input would introduce a self-consistency problem and would make the inferred relationship depend directly on the velocity used to evaluate it.

Recent machine-learning studies in glaciology have addressed related but distinct inverse and constitutive problems. Juvet (2023) inverted a convolutional emulator of Stokes flow, while Juvet and Cordonnier (2023) and Cheng and others (2024) incorporated ice-flow physics into learned forward or inverse models. Wang and others (2025) used remote sensing and physics-informed deep learning to infer spatially varying ice-shelf rheology. These approaches use spatial structure or governing equations to reproduce ice-flow fields or solve inverse problems. In contrast, the present study uses machine learning only as a flexible function approximator for $f(\cdot)$ in Eqn 1 to address a different question: can commonly available surface, bed, and geophysical observables contain sufficient information to define a transferable point-wise mapping for inversion-derived basal friction?

METHODS

We test this hypothesis using spatially withheld tests across the Amundsen Sea sector (Fig. 1). We evaluate the models in ten spatially separated 50 km squares, each with a surrounding buffer excluded from the training and validation datasets. A sector-wide basal-friction inversion, constructed using observed velocity from the complete sector, provides the reference C used to train the models outside each excluded region. The regression predicted friction field is then used in a forward ice-flow simulation within the withheld region, and the resulting modeled velocity is compared with observations. These experiments test whether a point-wise relationship fitted elsewhere in the sector remains useful in a withheld location. We compare six

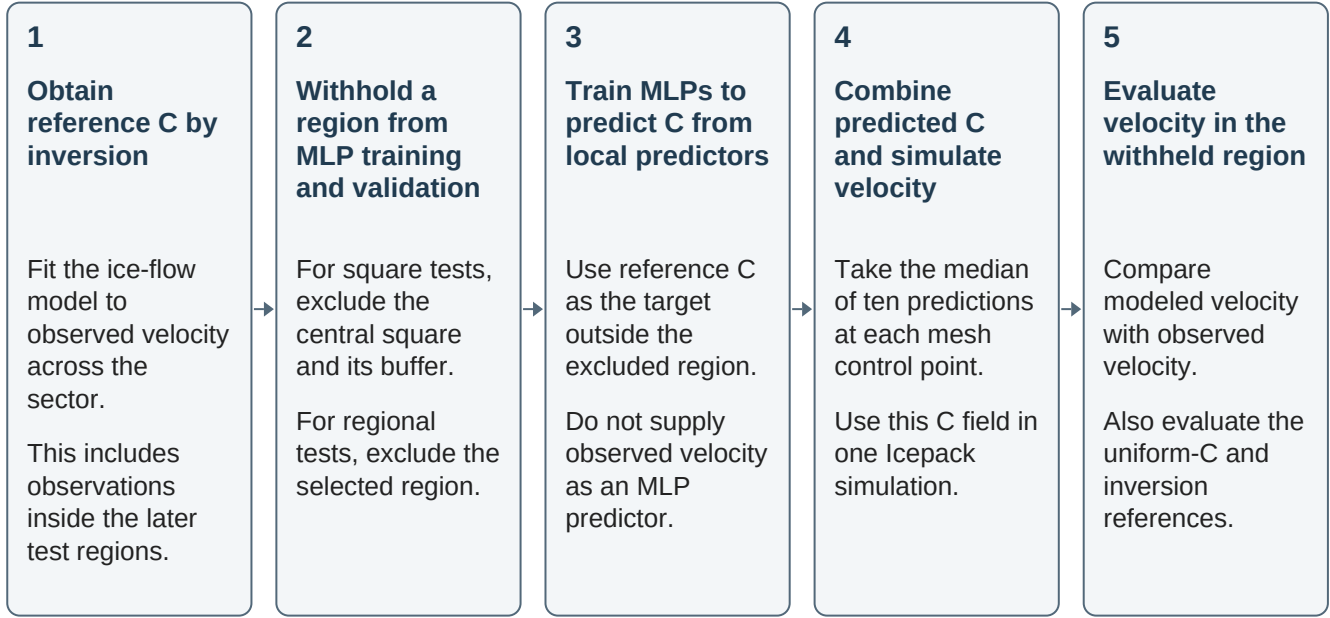


Fig. 1. Steps in the spatial-transfer experiment.

predictor configurations, quantify how well the training data represent the withheld predictor values, and perform a secondary catchment-scale test by withholding Pine Island Glacier. Because inversion-derived C is not necessarily unique (MacAyeal, 1993), we use modeled velocity as the primary evaluation, treating direct agreement with the reference C as secondary (Appendix section Basal-friction control: distributions and comparisons).

Study region and observations

The study domain covers the Pine Island, Thwaites, and Dotson–Crosson drainage systems in the Amundsen Sea sector of West Antarctica (Fig. 2). The catchment outlines were delineated manually, with lateral boundaries following glacier velocity flowlines. PIG, Thwaites, and Dotson are geographic labels; they do not imply that each catchment contains one basal regime. Two non-overlapping inter-catchment corridors complete the five-region division used for regional comparisons.

We take the velocity from the Making Earth System Data Records for Use in Research Environments (MEaSUREs) Phase-Based Antarctica Ice Velocity Map, derived from satellite radar interferometry (Mouginot and others, 2019). Surface elevation, ice thickness, and bed elevation are taken from BedMachine Antarctica, version 2 (Morlighem, 2020; Morlighem and others, 2020). BedMachine combines direct observations with reconstruction that uses ice velocity in some regions. The ice-product predictors therefore represent available geometry products, some of which already incorporate velocity information, although

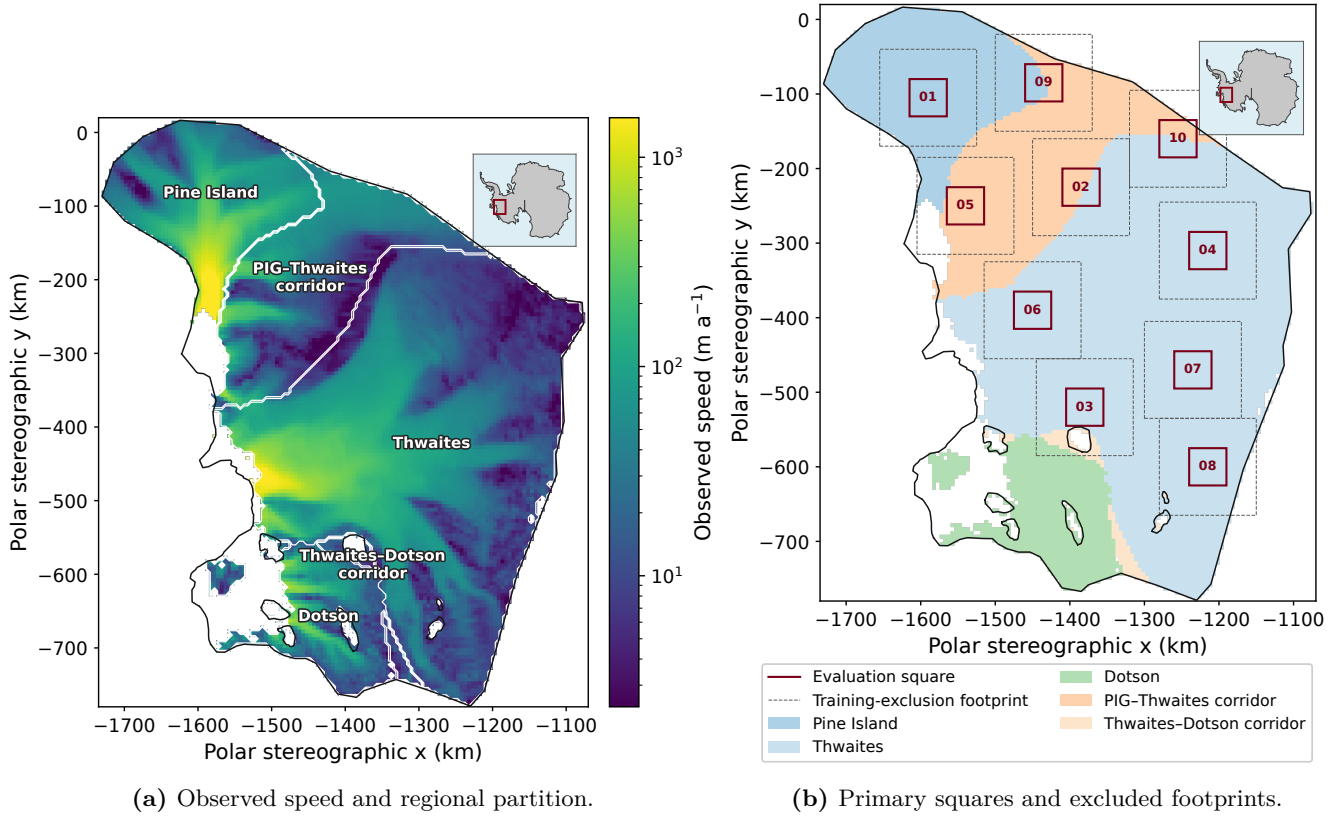


Fig. 2. Study region and holdout design. Panel (a) shows observed MEASUREs speed over the eligible part of the Amundsen Sea sector, with the PIG, Thwaites, and Dotson regions and the two inter-catchment corridors. Panel (b) shows the ten 50 km primary evaluation squares (solid) and their 130 km excluded footprints (dashed). Each footprint includes a 40 km buffer on every side of the central square and is excluded from training and validation datasets.

velocity itself is not used as a predictor. BedMachine uses different methods to reconstruct ice thickness between observations. Most of our test regions (detailed in the following sections) lie in regions mapped using streamline-diffusion interpolation, which favors smoother thickness variations along flow than across flow (Morlighem and others, 2020).

Surface air temperature and geothermal heat flux follow the products distributed with Quantarctica (Matsuoka and others, 2018; Comiso, 2000; An and others, 2015). The magnetic anomaly is taken from the 5 km EPSG:3031 ADMAP-2S grid (Eagles and others, 2024), and the gravity disturbance from the oriented EPSG:3031 AntGG2021 gravity-anomaly grid (Scheinert and others, 2024). Interpolation to the model grid adds no spatial information beyond what the original products contain, and their native resolutions and processing differ.

Ice-flow model and basal-friction inversion

All simulations used Icepack (Shapiro and others, 2021) and a depth-integrated stress balance based on the Shallow Shelf Approximation (SSA) (MacAyeal, 1989). The unstructured triangular mesh has 9,069 vertices and 17,653 elements over 310,546 km², with a median element edge length of 6.6 km. The basal-friction control uses continuous quadratic elements on this mesh, giving 35,797 control points. Because the shallow-shelf approximation neglects vertical shear, we compare the modeled horizontal velocity directly with the observed surface velocity. Because the inferred basal-friction control depends on the adopted stress-balance approximation and basal law, the model parameter C_b is model-dependent rather than a unique representation of basal conditions.

Basal friction is represented with a Weertman-type law,

$$\boldsymbol{\tau}_b = -C_b \|\mathbf{u}\|^{1/n-1} \mathbf{u}, \quad (2)$$

where $\boldsymbol{\tau}_b$ is the resisting basal shear stress on the ice, $\mathbf{u}$ is depth-averaged horizontal velocity, $n = 3$ is the exponent used in the basal law, and C_b is a dimensional basal-friction coefficient with units MPa (m a⁻¹)^{-1/3}. A Weertman law is a simplified hard-bed parameterization and does not explicitly represent processes such as effective-pressure dependence, cavitation, or deforming till. We hold the basal law fixed so that the experiment tests the proposed mapping with one consistent ice-flow model.

We prescribe observed velocities along the non-terminus boundaries and apply an ocean-pressure traction condition at the termini. Positivity and the grounding transition are imposed through

$$C_b = \phi C_0 e^C, \quad (3)$$

where C is the dimensionless inversion control used to represent spatial variation in basal friction and $C_0 = 0.01$ MPa (m a⁻¹)^{-1/3}. ϕ is a height-above-flotation ramp that continuously reduces basal resistance as flotation is approached and enforces $C_b = 0$ over floating ice (Leguy and others, 2014; Seroussi and others, 2014). Appendix section Ice-flow equations, grounding transition, and inversion method describes the full stress balance and grounding transition.

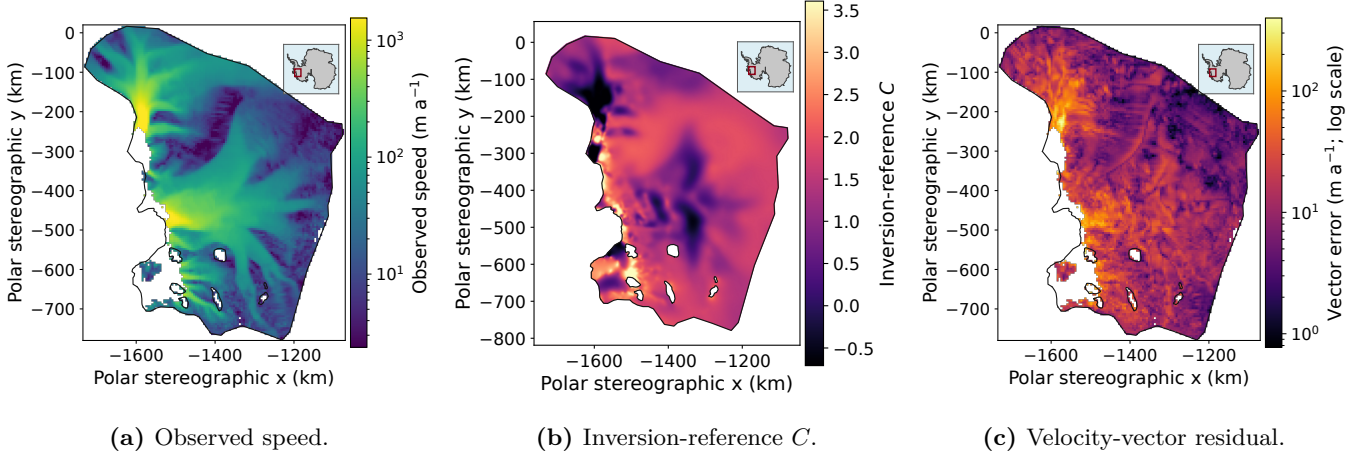


Fig. 3. Sector-wide inversion used to construct the learning target. The inversion field is one regularized solution for the stated data, model equations, sliding law, boundary conditions, and regularization, and provides the reference C for the learning experiments.

The inversion minimizes the total objective function

$$J(\mathbf{u}, C; r_C) = \frac{1}{2NU^2} \sum_{i=1}^N \|\mathbf{u}(\mathbf{x}_i) - \mathbf{u}_{\text{obs}}(\mathbf{x}_i)\|^2 + \frac{1}{2|\Omega|} \left(\frac{L}{r_C} \right)^2 \int_{\Omega} \|\nabla C\|^2 dA, \quad (4)$$

where N is the number of paired-velocity observations, $|\Omega|$ is the domain area, $U = 1 \text{ m a}^{-1}$ and $L = 7.5 \text{ km}$ are fixed reference scales that make both terms dimensionless, and r_C controls smoothness. Smaller r_C gives stronger smoothing. The misfit is unweighted, matching the velocity RMSE that we use for evaluation. Following the use of L-curves in ice-sheet inverse models (Wolovick and others, 2023), we selected $r_C \simeq 0.014$ using a maximum-curvature criterion (Appendix Fig. 8). We obtain gradients of Eqn 4 using the adjoint method. We use the resulting field (Fig. 3) for every learning experiment.

Dataset and predictor configurations

We first interpolate every predictor product to the control points of the model mesh, where the inversion also defines C , and compute the gradients and derived predictors from these finite-element fields. We then evaluate the fields at the 450 m MEaSURES pixel centers to construct the dataset, with each row containing the predictors and inversion-reference C at one pixel center. Because they are finite-element fields on the mesh, the predictors cannot resolve variation finer than the mesh: each central 50 km square contains 169–330 control points, whereas the 500 m BedMachine grid has about 10,000 cells in the same area. The trained MLPs are applied at the control points to produce the predicted C fields used in the forward

Table 1. Predictors in the six configurations. Ice-product predictors are denoted I and geophysical-product predictors G. A filled circle indicates inclusion.

Symbol	Predictor	Unit	Group	CFG01	CFG02	CFG03	CFG04	CFG05	CFG06
s	surface elevation	m	I	•	•			•	•
h	ice thickness	m	I	•	•			•	•
$\ \nabla s\ $	surface-gradient magnitude	1	I	•	•			•	•
$\ \nabla h\ $	thickness-gradient magnitude	1	I	•	•			•	•
τ_d	driving stress	MPa	I	•					
T_s	surface air temperature	°C	I	•	•			•	•
b	bed elevation	m	G			•	•	•	•
$\ \nabla b\ $	bed-gradient magnitude	1	G			•	•	•	•
Q_b	geothermal heat flux	mW m ⁻²	G			•	•	•	•
g_d	gravity disturbance	mGal	G			•			
m_a	magnetic anomaly	nT	G			•			
a_{bs}	bed–surface gradient alignment	1	G						•

simulations. Eligible rows must have finite values for all predictors and both velocity components, with $\phi > 0.1$ to exclude floating ice and the immediate grounding transition. We use the same eligibility criteria for all predictor configurations and experiments. Appendix Figure 9 shows this eligible region mapped against the full modeled domain. Table 1 summarizes the twelve predictors and their six configurations (CFG01–CFG06). Appendix section Eligible rows, predictor construction, and support provides gradient construction and exact row counts.

The one additional predictor in CFG06 relative to CFG05 measures bed–surface gradient alignment,

$$a_{bs} = \frac{\nabla b \cdot \nabla s}{\sqrt{(\|\nabla b\|^2 + \epsilon_b^2)(\|\nabla s\|^2 + \epsilon_s^2)}}, \quad (5)$$

where $\epsilon_b = 5.4 \times 10^{-3}$ and $\epsilon_s = 1.5 \times 10^{-4} \text{ m m}^{-1}$. When both gradients are large relative to these values, a_{bs} approaches $\cos \theta_{bs}$. The added terms prevent the alignment from becoming unstable where either gradient is very small. This predictor describes relative alignment without using velocity direction. The surface gradient defines the local driving-stress direction, so comparing it with the bed gradient measures whether bed slopes are oriented along or across the driving direction, without using observed velocity. The choice of ϵ_b and ϵ_s is described in Appendix section Eligible rows, predictor construction, and support.

CFG01 contains all ice-product predictors and CFG02 the selected ice-product subset. CFG03 contains

all geophysical-product predictors and CFG04 the selected geophysical-product subset. CFG05 combines the two selected subsets, and CFG06 adds Eqn 5. We selected the subsets used in CFG02 and CFG04 from preliminary catchment-scale analyses of modeled velocity conducted before the present spatially withheld experiments, and we did not modify them based on the square results.

Spatially withheld square experiments and predictor support

The primary experiment uses ten 50 km squares distributed across the study region (Fig. 2b). For each square, we exclude the full 130 km footprint from the training and validation datasets, leaving a 40 km buffer on each side of the central evaluation square. The ten excluded footprints do not overlap. Following the rationale of spatially separated evaluation (Roberts and others, 2017; Valavi and others, 2019), the buffer reduces dependence on nearby spatially correlated training rows. It does not remove the dependence introduced by the common sector-wide inversion or by non-local ice flow. The central 50 km square is the primary evaluation area; the surrounding buffer and complete footprint are used only for maps and secondary summaries, not as additional independent tests. The quantitative basis for the buffer is given in Appendix section Spatial separation and buffer choice.

We selected candidate square locations using spatial separation and predictor-support criteria so that the remaining training region generally represented the withheld predictor values well. We assessed support using both marginal and joint criteria. Marginal support requires every included predictor at a held-out point to lie within the 1st–99th percentile range of the corresponding eligible non-held-out rows. Joint support asks whether the complete predictor combination has a sufficiently close analog in the remaining training region. We report the area fraction satisfying both criteria and retain every held-out point in the modeled velocity evaluation regardless of support. Details of the joint-support transformation and distance threshold are given in Appendix section Eligible rows, predictor construction, and support. A separate regional experiment withholds the two inter-catchment corridors and is reported in Appendix section Basal-friction control: distributions and comparisons.

We used no criterion based on the inversion-reference C or modeled velocity errors to select squares. We examined the distribution of inversion-reference C only afterward because the reference field is a regularized, non-unique inversion solution, and a different predicted C may still reproduce the observed velocity. We did not use modeled velocity error to select the held-out regions, so we chose the test locations independently of their eventual forward-model performance.

Catchment-scale transfer test

In addition to the ten spatial tests, we retained Pine Island Glacier (PIG) as a catchment-scale transfer test chosen before we examined its results. This tests whether an MLP can transfer a point-wise relationship learned elsewhere in the Amundsen Sea sector across an entire glacier catchment. We excluded all eligible PIG rows from the training and validation dataset; the remaining grounded sector, including Thwaites, Dotson–Crosson, and the two inter-catchment corridors, provided the training and validation rows. We evaluated predictor support in PIG using the same marginal and joint criteria as the square experiments, without using C or velocity errors.

Model fitting and repeated realizations

We use an MLP as a flexible nonlinear representation of $f(\cdot)$ in Eqn 1 to test whether inversion-derived friction can be expressed as a local function of the available predictors. The model uses no information from neighboring locations; spatially structured approaches, such as convolutional or graph neural networks, would therefore test a different hypothesis. The MLP contains ten hidden layers of 200 units and predicts the reference C using the predictors in each configuration. We robustly scale inputs and targets using medians and interquartile ranges calculated from the training data only. Appendix section MLP training and statistical comparisons provides further details on the network implementation, training procedure, repeated realizations, and training and validation histories.

For each square experiment, we reproducibly split the eligible rows remaining after removing the excluded footprint into 90% for training and 10% for validation. Nearby training and validation rows are not statistically independent because of interpolation, predictor resolution, spatial autocorrelation, and regularization of the inversion-reference field. We therefore use the random split only for model fitting and selection, and evaluate performance separately in the spatially withheld central square. Appendix section Eligible rows, predictor construction, and support provides exact row counts.

For each experiment and predictor configuration, we train ten reproducible realizations that differ in the training/validation split, initial weights, and epoch shuffling. Using multiple realizations reduces sensitivity to any one set of training choices and provides a more stable estimate of the predicted C field. Matching realizations use the same row split across predictor configurations to support direct comparison. The spread among realizations reflects sensitivity to these training choices rather than calibrated predictive uncertainty or independent spatial variability. The primary prediction is the median of the ten predicted

C fields at each mesh control point, which is then evaluated in a separate forward simulation.

Forward simulations and measures of performance

For each experiment and predictor configuration, we evaluate the median of the ten predicted C fields, taken at each control point, in a separate Icepack forward simulation. For each square, predicted C replaces the inversion-reference C only within the complete 130 km excluded footprint, intersected with the common eligible grounded mask. The inversion-reference C is retained everywhere outside that footprint. The corresponding uniform- C simulation uses the identical replacement mask, so the two simulations differ only in the spatial pattern of C inside the excluded footprint. The PIG experiment uses the same procedure over the withheld catchment.

The primary performance measure is vector velocity RMSE (hereafter velocity RMSE) against the original paired MEaSUREs velocity components at their regular 450 m pixel centers over the evaluation area A . We evaluate modeled velocity at the same locations. The inversion misfit (Eqn 4) uses the same original pixel values. This evaluation is distinct from the mesh-interpolated observation field that the solver uses for boundary conditions and as its initial velocity. The spatial average below is approximated using the mean squared vector residual at these equally spaced points (Appendix section MLP training and statistical comparisons):

$$\text{RMSE}_u = \left[\frac{\int_A \|\mathbf{u}_{\text{ML}} - \mathbf{u}_{\text{obs}}\|^2 dA}{\int_A dA} \right]^{1/2}. \quad (6)$$

For the maps, we also use the local vector-error magnitude

$$e(\mathbf{x}) = \|\mathbf{u}(\mathbf{x}) - \mathbf{u}_{\text{obs}}(\mathbf{x})\|.$$

The spatially uniform- C reference provides a simple baseline using a single control value with no spatial variation from the predictors. This value is the finite-element-area-weighted mean inversion-reference C outside the withheld region and is applied uniformly within the withheld region. Because $C_b \propto \exp(C)$, averaging C corresponds to a geometric mean of $C_0 \exp(C)$ before multiplication by the grounding ramp ϕ . We do not fit the constant to held-out velocity; we apply it over the same withheld replacement region as the MLP-predicted field, retaining the inversion-reference C outside that region in both simulations. The comparison therefore asks whether spatial variation in C inferred from the predictors improves modeled velocity beyond what a single representative friction control provides. We also report $\text{RMSE}_{\text{relative}} =$

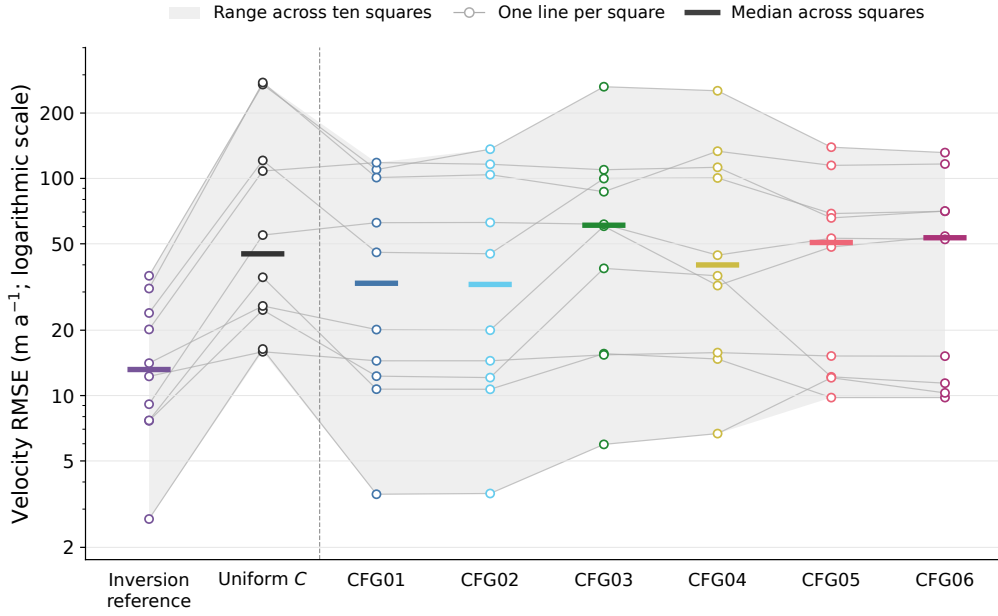


Fig. 4. Velocity RMSE against observed MEaSUREs velocity in the ten central squares. The inversion reference and square-specific uniform- C reference are followed by forward simulations using the median predicted C for all six configurations. Each gray line connects the same square; shading spans the minimum to maximum across squares, and short horizontal marks show the medians. The vertical axis is logarithmic. We calculate uniform C separately from the eligible region outside each excluded footprint; both the predicted and uniform controls replace the inversion reference only inside that footprint. The inversion reference was fitted using sector-wide observations, including those in the test squares.

RMSE_{ML}/RMSE_{uniform}; values below one indicate improvement over the uniform- C reference.

The ten held-out squares are the independent spatial tests. For each predictor configuration, we summarize performance by the median and interquartile range of the ten square RMSE values. Related configurations are also compared square by square to determine whether adding or removing predictors gives a consistent improvement across locations.

RESULTS AND DISCUSSION

Modeled velocity performance

Figure 4 shows the velocity RMSE for each held-out square, including the inversion and uniform- C references. The local observables often produce a C that improves modeled velocity relative to the spatially uniform- C reference, but performance varies substantially among locations. The two ice-product configurations have the lowest median RMSE values across the squares, 32.9 m a⁻¹ for CFG01 and 32.5 m a⁻¹ for CFG02, and each improves on uniform C in eight of ten squares (Table 2). CFG03 and CFG04, the all and selected-geophysical configurations, improve on uniform C in six and seven squares, respectively,

Table 2. Velocity RMSE against observed MEaSUREs velocity in the ten central squares. Values are the median and interquartile range across squares; the median RMSE in each row is taken over that row’s own ten squares and is not a paired comparison. Relative RMSE and the improvement counts compare each configuration with its square-specific uniform C reference within the same square. We fitted the inversion reference using sector-wide observations, including those inside the test squares; it therefore represents the fitted reference rather than a withheld prediction. RMSE is in m a^{-1} .

Control/config.	Predictors	Median RMSE	IQR	Better than uniform	Median relative RMSE	Relative RMSE IQR
Uniform C	spatially uniform	44.9	25.0–117.8	–	1.00	1.00–1.00
Inversion reference	–	13.2	8.0–23.0	10/10	0.19	0.17–0.29
CFG01	all ice	32.9	12.8–91.3	8/10	0.45	0.37–0.88
CFG02	selected ice	32.5	12.7–93.7	8/10	0.50	0.37–0.87
CFG03	all geophysical	60.9	21.3–96.6	6/10	0.97	0.54–1.09
CFG04	selected geophysical	40.0	19.9–109.5	7/10	0.88	0.56–1.03
CFG05	selected combined	50.7	13.0–68.1	9/10	0.59	0.50–0.90
CFG06	combined + alignment	53.3	12.4–70.7	9/10	0.61	0.47–0.88

while CFG05 and CFG06, which combine the selected ice-product and geophysical-product predictors, improve on uniform C in nine squares. The median RMSE is taken across the ten squares for each control separately, so the middle square for a configuration need not be the middle square for uniform C . The improvement counts and the relative RMSE compare the two controls within the same square. Because the squares differ widely in difficulty, CFG05 and CFG06 have a higher median RMSE than uniform C while still giving lower RMSE than uniform C in nine of the ten squares. In SQ04, for example, CFG05 lowers RMSE from 35.1 to 9.8 m a^{-1} ; that improvement moves SQ04 to the lowest value in the CFG05 distribution and therefore out of its median.

The inversion-reference solution has a substantially lower median RMSE of 13.2 m a^{-1} . This is not a withheld benchmark, because the inversion was fitted using velocity observations from the complete sector, including the evaluation squares. It nevertheless provides context for the velocity fit obtained with the spatially resolved inversion. The substantially larger errors from the predicted C fields show that the point-wise mappings recover only part of the improvement available from the inversion-derived spatial structure. At the same time, their frequent improvement over uniform C shows that the predictors contain information relevant to the effective friction control.

Removing driving stress from CFG01 to form CFG02 changes the median velocity RMSE only slightly, from 32.9 to 32.5 m a^{-1} , and both configurations improve on uniform C in eight squares. Because driving stress is derived from ice thickness and surface slope, which CFG02 already includes, supplying it explicitly

provides no consistent improvement across the ten squares. Among the geophysical configurations, CFG04 has a lower median velocity RMSE than CFG03 (40.0 versus 60.9 m a⁻¹), but CFG04 has lower velocity RMSE than CFG03 in only five of the ten squares; CFG03 performs better in the other five. Relative to uniform C , CFG03 reduces velocity RMSE in six squares and CFG04 in seven.

Combining the selected predictor groups does not consistently outperform either group alone. CFG05 performs better than CFG02 in three squares and better than CFG04 in seven. Adding bed–surface gradient alignment in CFG06 again divides the comparison evenly, with CFG06 performing better than CFG05 in five squares and worse in five. The paired sign tests provide no statistical evidence of a consistent advantage in these five configuration contrasts (Appendix section MLP training and statistical comparisons); this does not establish equivalence or negate the improvements over uniform C . Overall, the relative performance of the predictor configurations changes among squares, indicating that no tested predictor set provides a uniformly successful point-wise relationship across the study region.

Predictor support and modeled velocity performance

We next examine whether poor modeled velocity performance is associated with low predictor support. Across the ten primary squares, predictor support is generally high (Fig. 5a). Marginal support covers more than 94% of each primary square, while lower joint support is concentrated mainly in SQ08. However, high support does not guarantee good modeled velocity performance. Figure 5b shows performance relative to uniform C ; a relative RMSE greater than one means that the predicted control performs worse than the uniform- C reference. SQ07–CFG03, for example, has 99.4% support by both criteria but a relative RMSE of 2.33, while SQ02–CFG04 has 97.3% support and a relative RMSE of 1.44 (Table 3). These failures cannot be explained solely by individual predictors falling outside their training ranges or by the absence of a nearby multivariable match under the adopted definition.

Predictor support indicates whether similar predictor combinations are represented in the training region, but does not distinguish between densely and sparsely sampled parts of predictor space. We therefore examine whether more sparsely represented predictor combinations are associated with errors in the predicted C for CFG02 and CFG04 (Appendix section Training representation and prediction error). For CFG02, C RMSE increases with sparser representation in seven of the ten squares, but the overall relationship is weak. The median within-square Spearman correlation between training-representation density percentile and absolute C error is only 0.069, and the median C RMSE differs by only 0.004 across

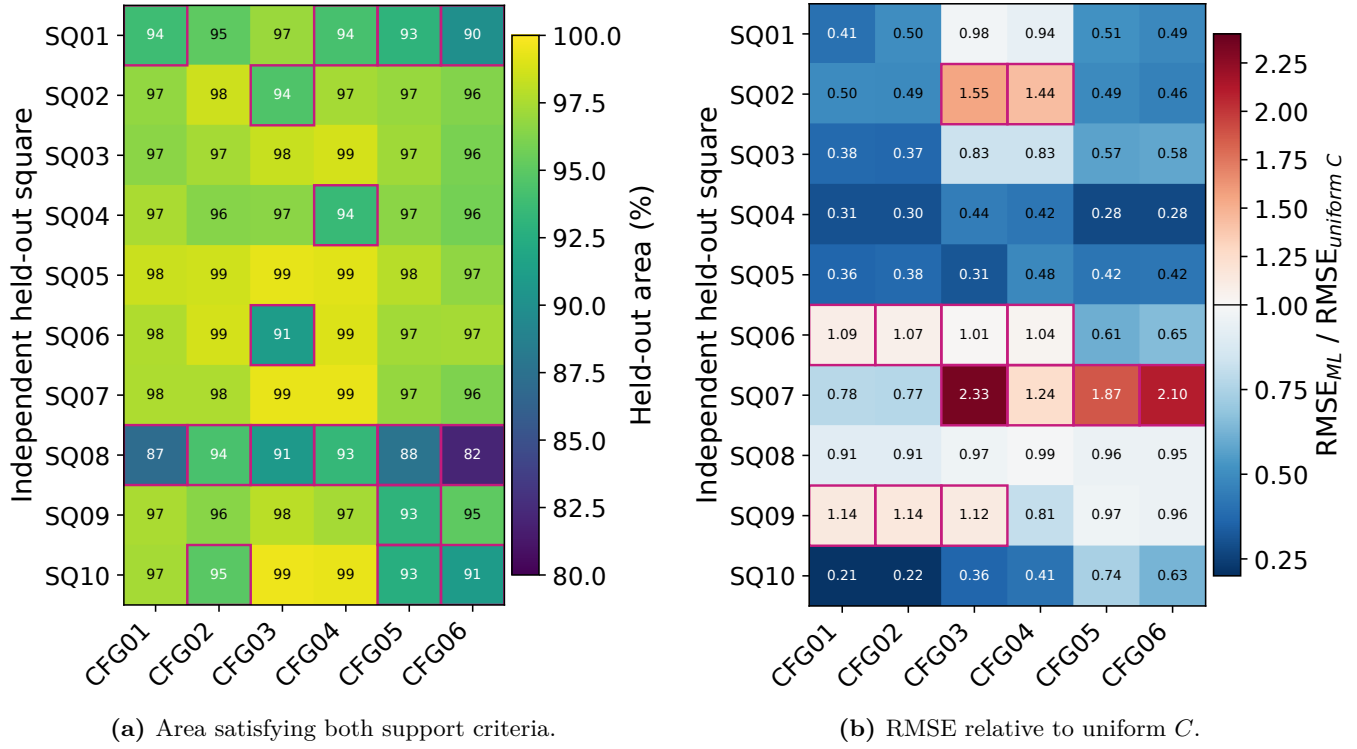


Fig. 5. Predictor support and modeled velocity performance in the ten spatial tests for six predictor configurations. Panel (a) gives the percentage of each held-out area satisfying both marginal and joint support (values rounded); magenta outlines mark values below 95%. Panel (b) gives relative RMSE; magenta outlines mark cases that do not improve on the spatially uniform- C reference. Marginal bounds use eligible non-held-out rows, while joint matches use the 5 km reference grid within the training region.

the three representation categories. CFG04 shows no similar pattern, with a median correlation between sparser representation percentile and absolute C error of -0.052 . Sparse training representation is therefore associated with larger C RMSE in some CFG02 experiments, but the relationship is weak and is not shared by CFG04.

Training representation also does not consistently correspond to modeled velocity performance, as illustrated by the square-level examples in Appendix section Training representation and prediction error. In some squares, modeled velocity has lower RMSE than uniform C in sparsely represented categories but not in better-represented categories. The same Appendix examples also show that closer agreement with the inversion-reference C does not consistently correspond to lower relative RMSE. Training representation therefore does not by itself explain the failures to improve modeled velocity relative to uniform C .

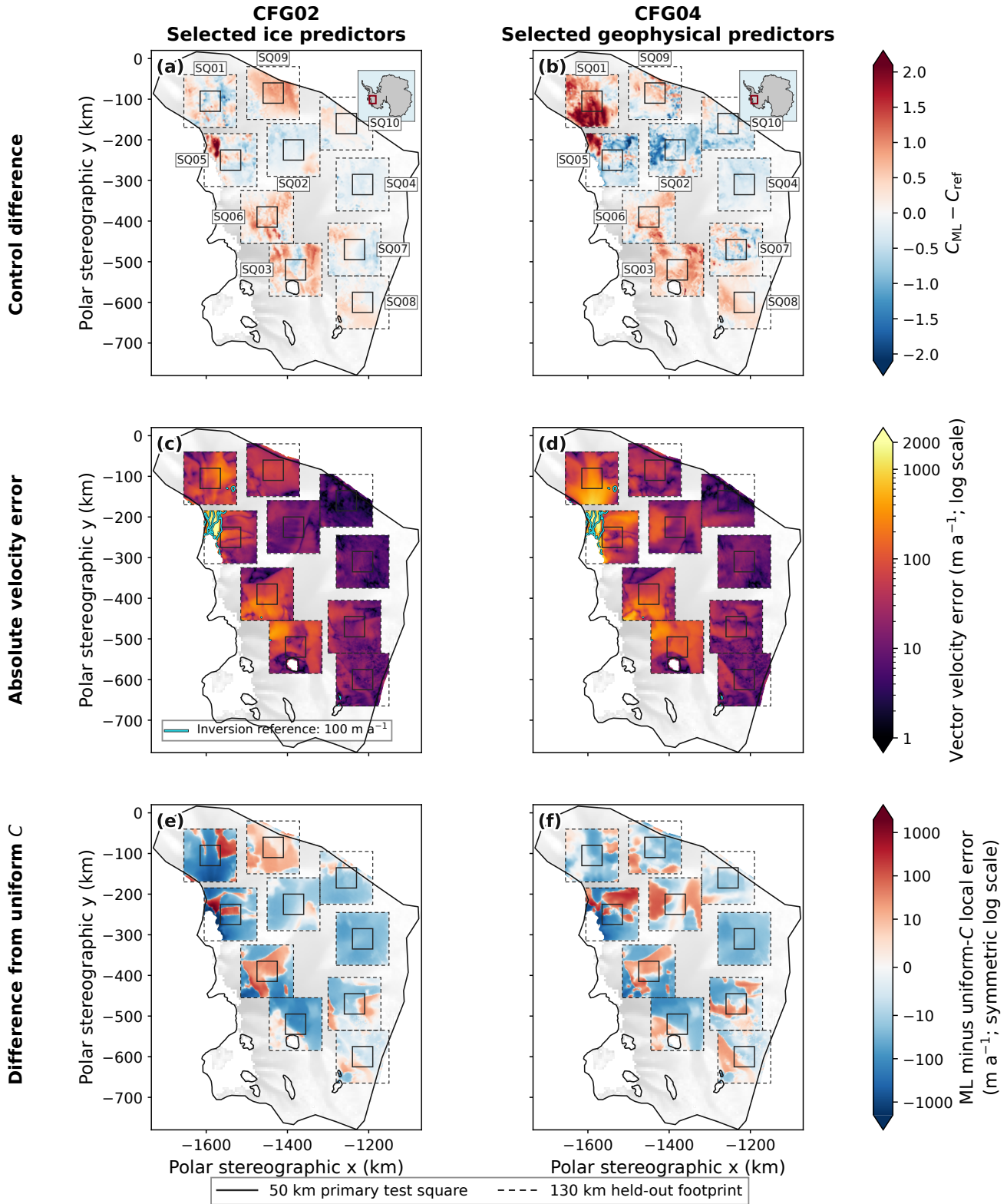


Fig. 6. Spatial comparison of the selected ice-product configuration (CFG02; left) and selected geophysical-product configuration (CFG04; right). Panels (a, b) show $C_{ML} - C_{ref}$. Panels (c, d) show the vector velocity error relative to MEaSUREs observations, $e_{ML} = \|\mathbf{u}_{ML} - \mathbf{u}_{obs}\|$. Panels (e, f) show $e_{ML} - e_{uniform}$; negative values indicate lower local error for the predicted control and positive values indicate lower error for uniform C . Cyan contours in panels (c, d) mark an inversion-reference velocity error of 100 m a^{-1} . In each experiment, both controls replace the inversion reference only inside the complete 130 km footprint. Velocity errors use the original paired MEaSUREs components and are shown as 1.8 km bin means for legibility; we calculated all statistics from the evaluated observation-grid points.

Table 3. Examples of poor performance, defined here as velocity RMSE exceeding the uniform- C reference, despite high predictor support. Support is the percentage of held-out points that satisfy both the marginal and joint criteria.

Square	Config.	Support (%)	ML RMSE	Uniform RMSE	Relative RMSE
(m a ⁻¹)					
SQ02	CFG04	97.3	35.6	24.8	1.44
SQ06	CFG01	97.6	118.3	108.2	1.09
SQ07	CFG03	99.4	60.2	25.9	2.33
SQ09	CFG02	96.4	62.7	54.8	1.14

Spatial patterns of success and failure

Figure 6 compares CFG02, the selected ice-product configuration, with CFG04, the selected geophysical-product configuration. The two ice-product configurations have nearly identical summary performance, so CFG02 represents that predictor group without repeating a visually similar map, and CFG04 represents the geophysical predictor group. For each square, we trained a separate set of MLPs, excluding the full 130 km footprint from training and validation; the panels therefore assemble ten separately withheld predictions rather than one sector-wide prediction. In every simulation, the predicted and uniform controls replace the inversion reference only inside the corresponding footprint. Across the central 50 km squares, CFG02 has lower velocity RMSE than CFG04 in eight of ten tests. Over the complete 130 km footprints, both configurations improve on uniform C in nine of ten tests, with median relative RMSEs of 0.64 for CFG02 and 0.86 for CFG04. Across the ten central 50 km squares, velocity error exceeds 100 m a⁻¹ over 13.0% for CFG02 and 22.8% for CFG04, compared with only 0.13% for the inversion-reference solution. Thus, although the predicted controls often improve on uniform C , regions of large absolute velocity error remain much more extensive than for the fitted inversion-reference control, as also observed in Table 2.

The maps show that the ice-product configuration provides an advantage across several distinct flow settings. In the fast PIG test SQ01, the central-square RMSE ratio is 0.50 for CFG02 and 0.94 for CFG04, with the largest CFG02 improvement concentrated in the fast-flowing part of the square. The local error difference adds an important distinction: CFG02 reduces error over 84.8% of the central square, with a median pointwise reduction of 133.5 m a⁻¹, whereas CFG04 reduces error at every evaluated point but has a much smaller median pointwise reduction of 14.8 m a⁻¹. Thus, the fraction of the square with lower local error does not determine the RMSE improvement; the magnitude and spatial distribution of the error reductions also matter. CFG02 also gives the larger improvement in moderately fast Thwaites flow in

SQ03 (0.37 versus 0.83). There, local error is lower than for uniform C over 90.4% of the central square for CFG02 and 77.5% for CFG04. Good performance is not confined to fast flow: both configurations improve substantially on uniform C in the slow Thwaites square SQ04 (0.30 and 0.42) and the slow PIG–Thwaites corridor square SQ10 (0.22 and 0.41).

The large absolute residuals in the SQ05 footprint occur where it extends from the PIG–Thwaites corridor into fast PIG flow. These large residuals contribute disproportionately to the footprint-wide RMSE, yet both learned controls still improve on uniform C (ratios of 0.85 and 0.86). Figure 6 panels (e, f) show that this footprint contains both reductions and increases in local error: CFG02 reduces error over 80.7% of its evaluated area and CFG04 over 60.2%. The large absolute errors in panels (c, d) therefore coexist with improvements relative to the uniform control. The inversion-reference velocity error is also elevated in this part of SQ05, indicating that some of the mismatch remains even with the fitted reference C . Over the complete footprint, however, the velocity RMSEs are 539.5 and 546.2 m a^{-1} for CFG02 and CFG04 respectively, compared with 77.1 m a^{-1} for the inversion reference. The predicted controls therefore introduce substantial additional error over the reference control.

Although CFG02 performs better than CFG04 in most squares, the relative performance of the two predictor groups varies by location. In SQ02, which spans the PIG–Thwaites corridor and Thwaites, CFG02 performs substantially better than the uniform- C reference, whereas CFG04 performs worse. The central-square RMSE ratios are 0.49 for CFG02 and 1.44 for CFG04, and the corresponding full-footprint ratios are 0.58 and 1.91. The local comparison shows that CFG02 reduces error over 99.8% of the central square, whereas CFG04 reduces it over only 42.5%. The higher CFG04 RMSE therefore accompanies a mixture of local improvements and deteriorations, rather than uniformly larger errors. CFG02 also agrees more closely with the inversion-reference control in the central square, with a C RMSE of 0.22 compared with 0.61 for CFG04. SQ09, spanning PIG and the PIG–Thwaites corridor, shows the opposite pattern: CFG04 improves on uniform C while CFG02 does not, with central-square ratios of 0.81 and 1.14 and full-footprint ratios of 0.88 and 1.20, respectively. In this central square, $e_{\text{ML}} - e_{\text{uniform}}$ is positive at every evaluated point for CFG02 and negative at every evaluated point for CFG04. CFG04 also has the lower central-square C RMSE in SQ09 (0.49 versus 0.74). These examples show that neither predictor group performs better at every location.

In SQ06, both configurations are slightly worse than uniform C in the central Thwaites square, with relative RMSEs of 1.07 for CFG02 and 1.04 for CFG04, despite simultaneous predictor support exceeding

98%. Local error is lower than uniform C over only 12.6% of the central square for CFG02 and 22.6% for CFG04. Over the complete footprint the RMSE ratios decrease to 0.62 and 0.79, and local error reductions cover 54.1% and 59.5% of the evaluated area. CFG04 changes in the same direction in SQ07, from 1.24 in the central square to 0.80 over the complete footprint. In other squares the result does not depend on the area evaluated: the CFG04 failure in SQ02 and the CFG02 failure in SQ09 remain over the complete footprints, with ratios of 1.91 and 1.20. The complete footprint includes the 40 km buffer, which lies closer to the training rows than the central square does, but the change between the two areas does not follow one direction: across the ten squares and these two configurations, the complete-footprint relative RMSE is lower than the central-square value in nine cases and higher in eleven. The complete footprint also covers a larger and more varied area than the central square, so the comparison depends on the ice included rather than on distance from the training rows.

SQ08 presents a different case: the uniform- C error is already only 15.9 m a^{-1} , compared with 12.3 m a^{-1} for the inversion-reference simulation, leaving relatively little scope for improvement. CFG02 and CFG04 give RMSE values of 14.5 and 15.8 m a^{-1} , respectively. Although local error is lower than for uniform C over 75.0% and 71.3% of the central square, the median local reductions are only 4.3 and 5.0 m a^{-1} . In this case, widespread negative values in panels (e, f) represent small absolute gains, and the larger residuals elsewhere limit the RMSE reduction.

As discussed above, agreement with the inversion-reference C and velocity performance coincide in SQ02 and SQ09. In fact, when comparing CFG02 with CFG04 within each of the ten central squares, the configuration with lower C RMSE also has lower velocity RMSE. However, the improvement in velocity RMSE from one configuration to the other does not follow directly from the magnitude of the control error. Over the complete SQ06 footprint, CFG02 and CFG04 have nearly identical C RMSE values (0.435 and 0.440), while their velocity RMSE ratios differ substantially (0.62 and 0.79). Conversely, over the complete SQ05 footprint, the lower C RMSE of CFG02 (0.57 versus 0.85 for CFG04) is accompanied by only a modest reduction in velocity RMSE (539.5 versus 546.2 m a^{-1}). Closer agreement with the inversion-reference control therefore does not translate into a proportional improvement in velocity RMSE. Within the individual squares, larger absolute local differences from the inversion-reference C generally coincide with larger local velocity errors, although the strength of this relationship varies by location. The median within-square Spearman correlations are 0.32 for CFG02 and 0.42 for CFG04, indicating a modest positive association rather than a one-to-one correspondence between C error and velocity error.

The spatial pattern of the control difference may help explain this variation. Membrane stresses couple neighboring locations, so a change in C can alter velocity beyond the location where it is applied. Local velocity error can therefore reflect control differences elsewhere as well as those at the same point. The sensitivity of the central-square results to the region over which C is replaced is quantified in Appendix section Sensitivity to the replacement region. Future work could use spatial sensitivity analysis to examine how perturbations to C in different parts of the domain contribute to the velocity response.

Taken together, the ten central-square tests show that the selected ice-product and geophysical-product configurations improve velocity RMSE relative to uniform C in most squares. Neither configuration improves every square, and some failures persist across the complete withheld footprint despite high predictor support. The selected predictors therefore contain information that can improve modeled velocity relative to a uniform control, but they do not define a point-wise relationship that is consistently successful across locations.

Catchment-scale test at Pine Island Glacier

The secondary catchment-scale experiment tests whether the point-wise relationship remains useful when all of PIG is withheld. We evaluate CFG02, the selected ice-product configuration, using the same ice-flow model and evaluation procedure as in the square experiments. The spatially uniform- C reference produces a vector velocity RMSE of 384.9 m a^{-1} , whereas CFG02 produces an RMSE of 354.8 m a^{-1} . This corresponds to a modest 8% reduction in RMSE relative to the uniform- C reference. The inversion-reference simulation has an RMSE of 48.1 m a^{-1} , showing that the transferred point-wise mapping recovers only a small fraction of the improvement obtained with the spatially resolved inversion.

CFG02 lowers the local velocity error relative to uniform C over 77.6% of the evaluated PIG area (Fig. 7). The improvement occurs across all observed-speed classes. The relative RMSE is 0.59 below 100 m a^{-1} , 0.84 between 100 and 500 m a^{-1} , 0.89 between 500 and 1000 m a^{-1} , and 0.96 above 1000 m a^{-1} . The improvement is therefore spatially widespread, but its influence on the catchment-wide statistic is concentrated in fast flow. The fastest 10% of the evaluated area, corresponding to observed speeds above approximately 504 m a^{-1} , contributes 59.8% of the total squared-error reduction relative to uniform C , but still accounts for 89.5% of the CFG02 squared error. In the fastest flow ($> 1000 \text{ m a}^{-1}$), the RMSE remains 1390.0 m a^{-1} , compared with 1448.8 m a^{-1} for uniform C . Thus, CFG02 improves much of the slow and intermediate-speed catchment interior and parts of the fast-flowing trunk, but provides only a modest

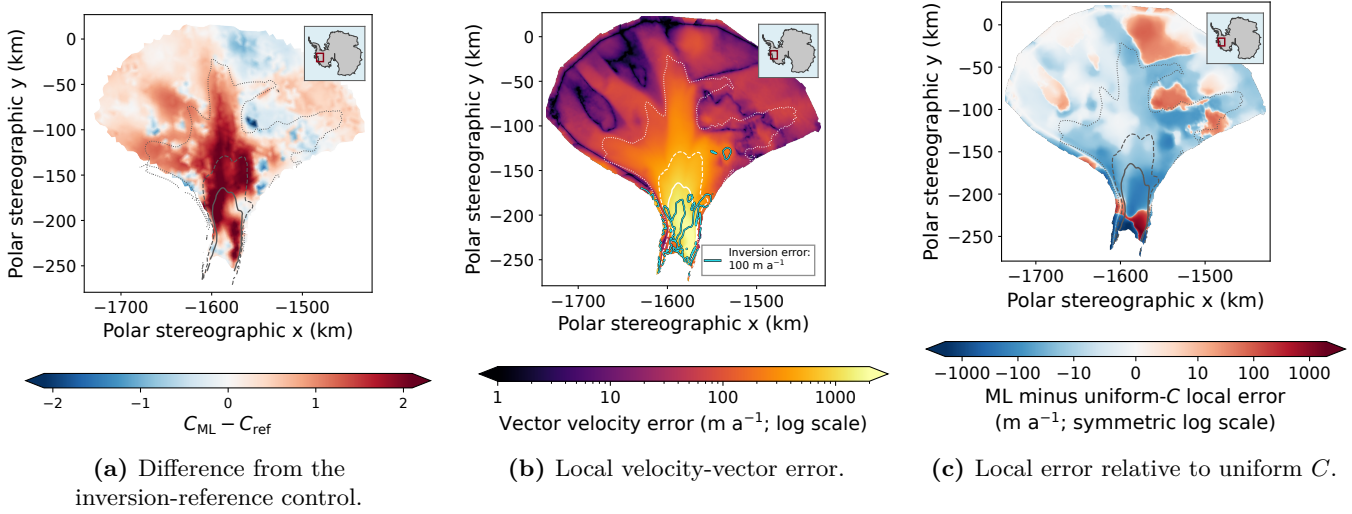


Fig. 7. Spatial evaluation of CFG02 with all of PIG withheld. Panel (a) shows the median predicted control relative to the inversion-reference control. Panel (b) shows $\|\mathbf{u}_{\text{ML}} - \mathbf{u}_{\text{obs}}\|$. Panel (c) shows the difference between the local CFG02 and uniform- C velocity errors; negative values indicate lower error for CFG02 and positive values indicate lower error for uniform C . Contours mark observed speeds of 100 (dotted), 500 (dashed), and 1000 m a^{-1} (solid). The cyan contour in panel (b) marks an inversion-reference velocity error of 100 m a^{-1} . Velocity fields are rendered from the original 450 m grid using display interpolation; all reported statistics use the original grid values. Both CFG02 and uniform C replace the inversion reference only inside the PIG holdout.

relative improvement in the fastest downstream flow. This spatial concentration explains how CFG02 can lower error across most of the catchment while retaining a large catchment-wide RMSE.

Locations where the inversion-reference error is at least 100 m a^{-1} comprise 3.9% of the evaluated PIG area but contain 47.1% of the CFG02 squared error. The remaining error is not confined to those locations: most locations with CFG02 error above 100 m a^{-1} lie outside regions where the inversion-reference error exceeds 100 m a^{-1} . Predictor support alone does not explain the spatial pattern. For CFG02, marginal support covers 97.5% of the evaluated PIG area, joint support covers 93.3%, and 91.9% satisfies both criteria. Of the net squared-error reduction relative to uniform C , 79.8% occurs within the area satisfying both support criteria. CFG02 nevertheless contains both successful and unsuccessful regions where predictor support is strong. CFG02 has a C RMSE of 0.85 across PIG. Locations with larger absolute differences from reference C also tend to have larger velocity errors, with a Spearman correlation of 0.67. This indicates a relatively strong positive association and is larger than the median within-square correlation of 0.32 for CFG02. The association between sparse training representation percentile and absolute C error is much weaker. C RMSE increases from 0.742 to 0.877 and 0.898 across the three categories of progressively sparser representation, but the Spearman correlation between representation percentile and absolute C error is only 0.031. This near-zero correlation indicates that fewer nearby training examples do

not consistently correspond to higher local C error across PIG.

Interpreting the point-wise relationship

Taken together, the experiments address three related questions: whether the predictor values in a held-out region are represented in the training data, whether the predicted C resembles the reference C , and whether the predicted C improves modeled velocity. The results show that these questions can have different answers. High predictor support can coexist with poor velocity performance, and sparser training representation does not consistently produce larger C errors. Agreement with the reference C is also related to, but does not determine, the resulting velocity error. Nor can the predictors, taken together, indicate in advance where the predicted C will improve on uniform C : a classifier trained on all twelve predictors is no better than predicting whichever outcome is more common in the withheld region, is not consistently better than one using observed speed alone, and does not transfer to PIG (Appendix Predicting where transfer succeeds). The PIG experiment shows that the same distinctions remain important when the held-out region is expanded from a 50 km square to an entire glacier catchment.

The contrast between random and spatial holdouts is itself a test of the hypothesis in Eqn 1. If C were a point-wise function of the predictors, a relationship that reproduces randomly withheld rows would hold equally in a spatially separate region wherever the predictor values are represented in the training data, and the two tests would agree. They do not. CFG02 reproduces randomly withheld validation rows with a median C RMSE of 0.08 and R_C^2 of 0.99 (Table 6), but in the withheld central squares its median C RMSE is 0.30 and R_C^2 is negative in eight of ten (Table 7), although 94–99% of every square satisfies both support criteria (Fig. 5a). Even held-out rows whose predictor combinations are at least as densely represented in the training data as a typical training row have a median C RMSE of 0.30 (Table 8), almost four times the validation value. The difference is therefore not explained by extrapolation beyond the training data or by sparse representation within it; it reflects structure in the inversion-derived C that the local predictors do not capture. Selecting the regularization or the stopping point against spatially withheld data might improve transfer, but under the point-wise hypothesis no such selection would be needed, and a gain obtained by smoothing the prediction toward a spatially uniform control would move away from, not toward, a point-wise relationship.

The spatial differences in performance are not simply explained by the BedMachine diagnostics considered here. Reported bed-elevation uncertainty has only a weak relationship with local velocity error across

the square tests (Appendix BedMachine reconstruction and velocity error). Comparisons among BedMachine source classes are more limited because eight of the ten central squares contain only streamline-diffusion cells. Nevertheless, CFG02 velocity RMSE ranges from 3.5 to 136.3 m a⁻¹ across those eight squares, showing that differences in reconstruction class alone cannot explain the large spatial variation in performance. Reconstruction may still matter within a source class; for example, smoothing may remove local bed variations that could help distinguish basal settings, but the present analysis does not isolate this effect.

More fundamentally, the available predictors may not capture all of the physical variability relevant to C . For example, Hoffman and others (2022) showed that small-scale basal roughness can strongly influence basal resistance, but continental-scale datasets do not resolve these scales. The geophysical products also provide indirect information at relatively coarse spatial resolution. In addition, the inferred C depends on the adopted stress balance, basal law, and regularized inversion, rather than representing a unique physical property of the bed.

The point-wise formulation itself may also be limiting. The MLP uses only the predictor values at each location and cannot use information from neighboring locations. At the same time, membrane stresses couple the velocity response across space, so changes in C at one location can affect velocity elsewhere. This is consistent with theoretical limits on local basal inference, in which basal disturbances are filtered through spatially coupled ice flow before appearing at the surface (Gudmundsson, 2003; Raymond and Gudmundsson, 2005; Gudmundsson and Raymond, 2008). The present experiments therefore identify several possible explanations for the remaining errors, but do not determine which mechanism is responsible in any individual region.

CONCLUSION

This study tested whether inversion-derived basal-friction control C can be represented by a point-wise relationship with commonly available surface, bed, and geophysical observables. We evaluated this relationship in ten spatially held-out regions across the Amundsen Sea sector and in a separate PIG catchment test by inserting the predicted C fields into Icepack and comparing the resulting velocities with observations.

The tested local observables contain information that can produce C fields that improve modeled velocity relative to uniform C in held-out regions, but that relationship is not spatially consistent across the Amundsen Sea sector. The ice-product configurations give the lowest median velocity errors, and the

combined configurations improve on a spatially uniform- C reference in nine of ten squares. However, performance varies strongly across locations; adding predictors does not systematically improve performance, and poor performance can occur even when the training data represent the held-out predictor values well. Training representation provides no simple explanation either: sparser representation does not consistently correspond to larger C errors across the square tests. Withholding all of PIG shows the same limitation at catchment scale, with CFG02 reducing velocity RMSE by only about 8% relative to uniform C . Although CFG02 lowers local velocity error over most of PIG, the remaining error is strongly concentrated in fast flow.

The spatially withheld squares show that high predictor support can coexist with poor velocity performance. Agreement with the reference C is also only partly related to velocity performance. Local C error and velocity error are positively associated, but the relationship is not one-to-one, and closer agreement with C does not translate proportionally into better velocity performance. Under the framework considered here, the Amundsen Sea sector, the selected data products, a point-wise MLP, and a shallow-shelf/Weertman inversion, the tested surface, bed, and geophysical observables yield fitted controls that improve modeled velocity in most test regions. More broadly, the results show that researchers should test such parameterizations in spatially held-out regions and evaluate them through modeled velocity, rather than only through agreement with reference C . Future studies can use the same framework to test other predictors and methods that use information from neighboring locations. The tested local observables do not support a single point-wise relationship with inversion-derived C that is consistently successful across locations.

CODE AND DATA AVAILABILITY

The inversion, velocity-simulation, and model-training code is available at <https://github.com/racheetmatai/icepack-inverse>. The reproduction artifacts, comprising the scientific inputs, the frozen experiment design, the canonical dataset and exact splits, the trained models, the friction-control fields and forward-simulation outputs, and the figure sources, are archived at Zenodo (<https://doi.org/10.5281/zenodo.22839669>). BedMachine Antarctica and the MEaSUREs velocity mosaic are not redistributed there; they remain available from NSIDC and are identified by exact checksums in the code repository.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

STATEMENT ON THE USE OF AI

We used generative AI tools to assist with language revision and consistency checking. The authors reviewed all scientific decisions, analyses, interpretations, and the final text.

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ICE-FLOW EQUATIONS, GROUNDING TRANSITION, AND INVERSION METHOD

Icepack solves the depth-integrated shallow-shelf stress balance

$$\nabla \cdot (h\mathbf{M}) + \boldsymbol{\tau}_b = \rho_i g h \nabla s, \quad (7)$$

where h is ice thickness, s is surface elevation, ρ_i is ice density, g is gravitational acceleration, and $\boldsymbol{\tau}_b$ is the basal traction acting on the ice. For horizontal depth-averaged velocity $\mathbf{u}$, the membrane-stress tensor used here is

$$\mathbf{M} = 2\mu [\dot{\boldsymbol{\epsilon}} + \text{tr}(\dot{\boldsymbol{\epsilon}})\mathbf{I}], \quad \mu = \frac{1}{2} A^{-1/n} \dot{\epsilon}_e^{1/n-1}, \quad (8)$$

with $\dot{\epsilon} = (\nabla \mathbf{u} + \nabla \mathbf{u}^T)/2$, $\dot{\epsilon}_e = [(\dot{\epsilon} : \dot{\epsilon} + \text{tr}(\dot{\epsilon})^2 + \dot{\epsilon}_{\min}^2)/2]^{1/2}$, $\mathbf{I}$ the horizontal identity tensor, $\dot{\epsilon}_{\min} = 10^{-5} \text{ a}^{-1}$ Icepack’s default strain-rate floor, A the temperature-dependent rate factor, and $n = 3$ the Glen exponent. The modeled velocity is depth-averaged. Its comparison with the MEaSUREs surface velocity therefore uses the shallow-shelf assumption that vertical shear is negligible. We use the depth-averaged englacial temperature of Van Liefferinge and Pattyn (2013). We linearly interpolate this field to the model mesh and supply it in kelvin to Icepack’s rate-factor function to calculate A (Shapiro and others, 2021). This differs from the surface-air-temperature predictor used by the MLP.

The basal law is Eqn 2, with the Icepack sliding exponent $n = 3$ in that equation. The dimensionless basal-friction control C is distinct from the dimensional basal-friction coefficient C_b that enters the traction law:

$$\begin{aligned}
 p_w &= \rho_w g \max(0, h - s), & p_i &= \rho_i g h, \\
 \phi &= \begin{cases} 0, & p_i = 0, \\ \max[1 - (p_w/p_i)^p, 0], & p_i > 0, \end{cases} \\
 C_b &= \phi C_0 \exp(C).
 \end{aligned} \tag{9}$$

Here ρ_w is seawater density, p_w is the hydrostatic water-pressure estimate, p_i is ice overburden pressure, $p = 1$ is the ramp exponent used in this study, and $C_0 = 0.01 \text{ MPa (m a}^{-1})^{-1/3}$. Consequently C_b has units $\text{MPa (m a}^{-1})^{-1/3}$, while C and ϕ are dimensionless. The ramp is continuous and piecewise, with kinks introduced by the maximum operations; it is not a differentiable smooth effective-pressure law.

Quadratic finite-element basis functions are used. We prescribe observed velocities along the non-terminus boundaries and apply an ocean-pressure traction condition at the termini. In that terminus condition the water-depth magnitude is $d = \max(h - s, 0)$ and the normal-stress contribution is proportional to $(\rho_i g h^2 - \rho_w g d^2)/2$. We filled eleven Dirichlet velocity degrees of freedom that lacked a colocated valid velocity pixel from the nearest valid centered MEaSUREs pixel.

For each square, predicted C is applied only within the complete 130 km excluded footprint intersected with the common eligible grounded area where $\phi > 0.1$. The PIG experiment uses the corresponding catchment mask. The uniform- C simulations use identical masks. Elsewhere, including across the grounding transition, we retain the reference C . This keeps the geometry, boundary conditions, and grounding treatment identical while holding the surrounding reference control fixed.

Sensitivity to the replacement region

We also compared the controlled design with an earlier calculation in which the predicted C , or the corresponding uniform- C value, replaced the inversion-reference control throughout the common eligible grounded sector. In that earlier design, the predicted control was therefore also used in the MLP training geography. In the controlled design used for the main results, replacement is restricted to the complete 130 km footprint for each square, or to the held-out regional geography, intersected with the same common eligible grounded mask. Predicted and uniform controls use identical masks, and both retain inversion-reference C outside the mask. The complete 40 km buffer around each central square is part of the replacement region.

We evaluated the effects of correcting the observational reference and changing the replacement geography separately. Replacing the mesh-interpolated observational values with the original paired MEaSUREs components changed the central-square MLP RMSE by a median of 0.07 m a^{-1} and a maximum of 0.95 m a^{-1} . Restricting replacement to the withheld geography then changed the central-square MLP RMSE by a median of 0.003 m a^{-1} and a maximum of 2.02 m a^{-1} . Neither change altered whether a configuration improved on uniform C in any square or which configuration had the lowest RMSE in a square. Differences were larger over the complete footprints. For CFG02 and CFG04, the median relative RMSE changed from 0.54 and 0.82 under sector-wide replacement to 0.64 and 0.86 under controlled replacement; both configurations improved on uniform C in nine of ten complete footprints in both designs. Across these 20 footprint comparisons, the median absolute change in relative RMSE was 0.04 and the largest was 0.28. The principal central-square findings are therefore robust to the two replacement designs, while the complete-footprint results are more sensitive to the surrounding control prescription.

The basal-friction control uses continuous quadratic finite elements. The replacement mask selects values at the finite-element control points, and outside-mask control-point values equal the inversion-reference C exactly. Within cells crossed by the geographic boundary, the quadratic basis connects inside- and outside-mask values. The transition can therefore be steep but is not a literal discontinuity in C , and we applied no explicit taper. The central square is separated from the footprint boundary by the 40 km buffer. All 76 controlled solves converged at the primary limit of 50 Newton iterations, with no retries at the higher limit, but convergence and this separation do not show that boundary influence is negligible. The change in modeled velocity between the two designs is consistent with spatial coupling in the ice-flow model, but it does not isolate a uniquely non-local contribution because the surrounding

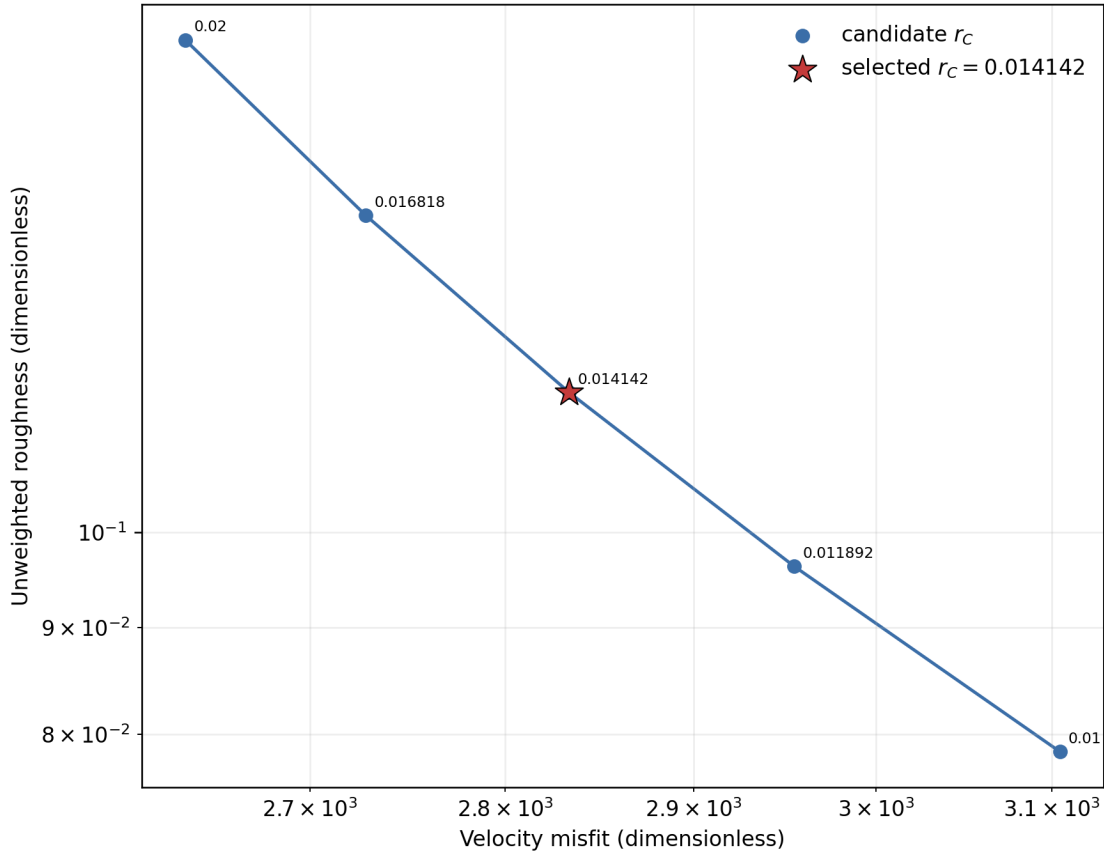


Fig. 8. L-curve for the sector-wide inversion. The axes are the two dimensionless terms of the objective in Eqn 4: the velocity misfit and the smoothness term without its factor r_C^{-2} . The maximum-curvature criterion selects $r_C \simeq 0.014$; smaller r_C corresponds to stronger smoothing in the objective function in Eqn 4. The neighboring candidate near $r_C \simeq 0.017$ has similar curvature, indicating a narrow elbow range.

control prescription and the transition at the replacement boundary change together.

We initialized the inversion with $C = 0$ and ran it for 300 optimization iterations. At that point the gradient norm was 6.7×10^{-5} , below the 10^{-3} ceiling adopted for acceptance, and the relative changes in misfit and in unweighted roughness over the final iterations were 1.4×10^{-5} and 3.4×10^{-5} . Figure 8 shows the L-curve used to select the regularization parameter. The maximum-curvature criterion selected $r_C \simeq 0.014$ from the refined candidates. The adjacent candidate, $r_C \simeq 0.017$, had curvature within approximately 4% of the selected value, indicating a narrow elbow range rather than a uniquely resolved regularization value. We also tried values of r_C outside this range. With weaker smoothing, $r_C = 0.05$ did not meet the convergence criteria within the iteration limit, and $r_C = 0.1$ failed numerically. The one value tried with stronger smoothing, $r_C = 0.005$, came from a preliminary run. We therefore do not show these values.

SPATIAL SEPARATION AND BUFFER CHOICE

For each 50 km evaluation square, we exclude an additional 40 km buffer from MLP training and validation on all sides, giving a 130 km excluded footprint. The buffer is intended to reduce the influence of nearby, spatially correlated training data on the withheld-square test. We chose the 40 km distance before model fitting using the spatial scale of observed ice-speed variations.

Using MEaSUREs speed v , we formed

$$q(\mathbf{x}) = \ln[1 + v(\mathbf{x})/v_0], \quad v_0 = 1 \text{ m a}^{-1}. \quad (10)$$

The logarithm reduces the influence of the fastest outlets, while the offset keeps the transformation finite where speed approaches zero. We then removed the broad sector-scale variation using a normalized Gaussian average with a 50 km standard deviation, leaving local variations in log speed. These local variations have an estimated $1/e$ correlation length of 22.6 km, and their correlation falls to 0.108 at a separation of 40 km. Varying the Gaussian scale from 30 to 100 km yields $1/e$ correlation lengths of 16.1–46.0 km.

An independent estimate from fast-tributary widths gave a 90th-percentile width of approximately 36–38 km across speed thresholds of 100, 250, and 500 m a⁻¹. The 50 km test side therefore spans the coarsest 5 km predictors ten times and exceeds this tributary-width estimate. A 40 km buffer reduces the local log-speed-anomaly correlation to 0.108. Increasing the buffer to 50 km reduces it only to 0.072 while increasing each excluded footprint from 16,900 to 22,500 km², a 33% increase. We therefore selected 40 km to balance separation from training rows against the loss of training and validation data. The buffer separates the test square from nearby training locations at a distance where departures from the broad observed-speed trend have a much weaker spatial correlation than at nearby locations. The sector-wide inversion and non-local membrane-stress response still couple the withheld square to the surrounding ice.

ELIGIBLE ROWS, PREDICTOR CONSTRUCTION, AND SUPPORT

This section defines the common set of observation-grid rows used across all predictor configurations, describes how the derived predictors are constructed, and explains how predictor support is evaluated in the held-out regions.

We begin by defining the rows eligible for model training and evaluation. The MEaSUREs velocity raster subset covering the study region contains 2,640,250 pixels. Requiring both observed velocity components

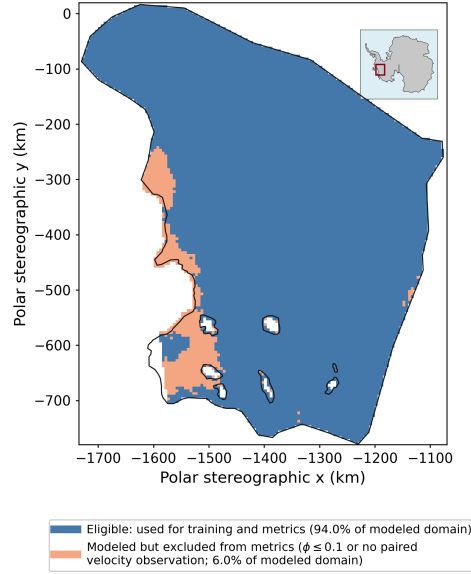


Fig. 9. The modeled domain and the eligible region used for training and for every reported metric. The excluded fringe (orange, 6.0% of the modeled domain by 5 km grid cell) is where $\phi \leq 0.1$ or no paired MEaSUREs velocity observation is available; it is computed and simulated but never scored. We show the two together here because every other map in this paper already restricts display to the eligible region alone.

to be finite and the MEaSUREs source flag to be positive retains 2,294,138 paired-velocity pixels; 1,622,598 of these lie inside the computational mesh. All twelve predictors are finite at those mesh-contained rows. Applying $\phi > 0.1$ leaves 1,530,992 common eligible rows, which are used by every predictor configuration. No additional rows are removed for individual configurations. Figure 9 maps this eligible region against the complete modeled domain. The inversion-reference C is interpolated from the model mesh to these observation-grid locations.

For the ten square experiments, excluding the 130 km footprint leaves 1,447,471–1,463,861 rows available for training and validation. Each MLP’s reproducible 90/10 split contains 1,302,723–1,317,474 training rows and 144,748–146,387 validation rows. The central-square evaluation sets contain 12,321 rows, except SQ03, which contains 12,432. For each training realization, every predictor configuration uses the same training locations and the same validation locations. Rows outside the common grounded eligible area retain inversion-reference C in every forward simulation. White or uncolored regions in maps denote locations outside the area shown for that experiment or outside that common mask; they were not removed on the basis of target values or model performance.

Surface elevation s , thickness h , and bed elevation b are represented as finite-element scalar fields. Their gradients are evaluated as finite-element vector expressions, interpolated component-wise to the observation

grid, and summarized as $\|\nabla f\| = (f_x^2 + f_y^2)^{1/2}$ for $f \in \{s, h, b\}$. Driving-stress magnitude is

$$\tau_d = \rho_i g h \|\nabla s\|, \quad (11)$$

reported in the stress unit used by Icepack, MPa.

Eqn 5 gives the alignment predictor. The small-gradient scales were estimated by dividing representative elevation-error scales by the median mesh-edge length of 6553.4 m. For the surface, a 1 m elevation-error scale based on the reported REMA accuracy gives $\epsilon_s = 1.5 \times 10^{-4} \text{ m m}^{-1}$. For the bed, the median BedMachine bed-elevation error over grounded mesh points is 35.31 m, giving $\epsilon_b = 5.4 \times 10^{-3} \text{ m m}^{-1}$. We rounded and fixed the resulting values before evaluating the model. We clip the alignment value to $[-1, 1]$. The ϵ_b and ϵ_s terms in the denominator reduce the magnitude of the alignment toward zero when either gradient is small relative to its corresponding scale.

We took the selected ice-product and geophysical-product subsets from an earlier catchment-level comparison of modeled velocity and specified them before examining the present spatial results. Table 4 summarizes the native spatial resolution and interpretation of the products used. Interpolating to the observation grid increases the number of sampled values but adds no new spatial information.

We evaluate marginal support separately for each predictor. A held-out point satisfies the marginal criterion when every predictor included in that configuration lies within the 1st–99th percentile range of the corresponding eligible non-held-out rows.

Joint support evaluates the full combination of predictor values. We first convert predictor values on a fixed 5 km grid covering the sector to rank-based normal scores by matching their ranks to standard-normal quantiles (Beasley and others, 2009). We then use principal component analysis (PCA; Jolliffe and Cadima, 2016) to reduce the predictor space while retaining at least 99% of the variance of the transformed predictors, and scale the retained components to unit variance. We use the same transformation for all spatial experiments.

To define the joint-support criterion, we use this grid to establish what counts as a sufficiently similar predictor combination. For each grid point, we calculate the distance in transformed predictor space to the most similar point located at least 40 km away along at least one grid axis, so that the excluded neighborhood is a square, matching the square shape of the 40 km buffer. The 95th percentile of these distances defines the joint-support threshold. For each held-out point, we then find the most similar 5 km reference-grid point within the corresponding training region. The held-out point satisfies the joint criterion

Table 4. Sources and source-grid spacing of the predictors. Nominal cells in a 50 km square are geometric counts of the source grid before masks and are not necessarily independent samples. All predictors are interpolated to the model mesh before use, and a central square contains 169–330 mesh control points, so the BedMachine-derived predictors retain only the variation resolved by the mesh. Derived gradients retain the resolution limits of their source fields.

Predictor(s)	Product/source	Source grid	Nominal cells	Interpretation in this study
s, h, b and their gradients	BedMachine Antarctica v2	500 m	$\sim 10,000$	Geometry products. Fast-flow bed reconstruction uses mass conservation and velocity, and is not a direct measurement of local basal processes.
τ_d	Derived from BedMachine h, s	500 m source fields	$\sim 10,000$	Large-scale driving-stress estimate derived from thickness and surface slope.
T_s	Quantarctica surface- air-temperature raster	5 km	~ 100	Large-scale surface thermal boundary, not a direct observation of basal temperature.
Q_b	Quantarctica geothermal-heat-flux raster	5 km	~ 100	Model-derived regional basal heat-flux estimate; interpolation does not add fine-scale information.
m_a	ADMAP-2S EPSG:3031 grid	5 km	~ 100	Magnetic anomaly containing broad information about crustal magnetization; it does not uniquely identify basal material.
g_d	AntGG2021 oriented EPSG:3031 grid	5 km	~ 100	Gravity disturbance containing information about subsurface density; it does not uniquely identify rock type or subglacial water.
a_{bs}	Derived from BedMachine b, s	500 m source fields	$\sim 10,000$	Relative alignment of two geometry gradients; does not use flow direction.

when this distance does not exceed the threshold.

We apply the same construction at two stages, using different predictor sets. Square selection used all twelve predictors, so a candidate square had to be supported in the full predictor space: at least 95% of its area had to lie within the marginal bounds of each predictor, and at least 95% had to satisfy the joint criterion. This is a stricter requirement than the one applied to any individual configuration. The support values reported for each configuration use only that configuration’s predictors and therefore have their own thresholds.

Reported predictor support is the area fraction satisfying both the marginal and joint criteria. We retain all held-out points in the modeled velocity evaluation, regardless of support.

BASAL-FRICTION CONTROL: DISTRIBUTIONS AND COMPARISONS

We first check whether the reference C values in each held-out region fall within the range of values available in the training-and-validation region. This is separate from predictor support, which considers only the input variables. We did not use reference C to select the square locations, define predictor support, or remove evaluation rows.

Figures 10, 11 and Table 5 compare the reference C distributions in the training-and-validation and held-out regions. All predictor configurations within a given experiment use the same combined training-and-validation and held-out rows, so the reference- C distributions are also the same for all configurations in that experiment.

These comparisons show how the reference- C range in each held-out region overlaps that in the training-and-validation region; modeled velocity tests whether the predicted control reproduces the observed flow. The reference C is one regularized, model-dependent inversion solution, and modeled velocity remains the primary evaluation in this study.

We measure agreement with the inversion-reference control using C RMSE and the coefficient of determination R_C^2 . The latter is one minus the sum of squared prediction errors divided by the sum of squared deviations of reference C from its mean over the same evaluated locations.

Table 6 reports agreement with the inversion-reference control on the validation rows. Each MLP was evaluated on its own validation rows, so the table reports the median of the ten member-level results for each square-configuration experiment. Validation C RMSE is lower and R_C^2 higher for the other five configurations than for CFG04. The contrast with the spatially held-out results in Table 7 shows that

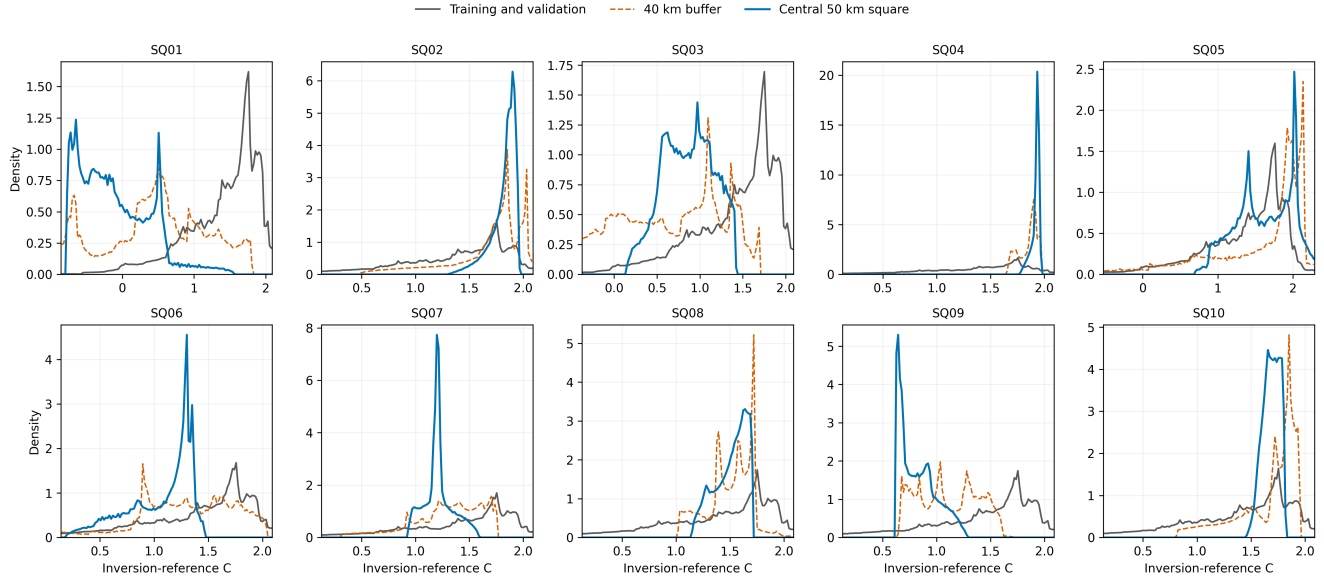


Fig. 10. Reference- C distributions for the ten withheld squares. Each panel compares the training and validation rows with the central 50 km evaluation square and the surrounding buffer. We choose the horizontal range separately for each panel to show the central part of the distributions clearly. Each density is normalized using all rows in its set, including values outside the displayed range. Numerical 1st–99th-percentile intervals and the percentage of held-out values within the training-and-validation interval are reported in Table 5. Reference C was not used to select square locations or remove evaluation rows.

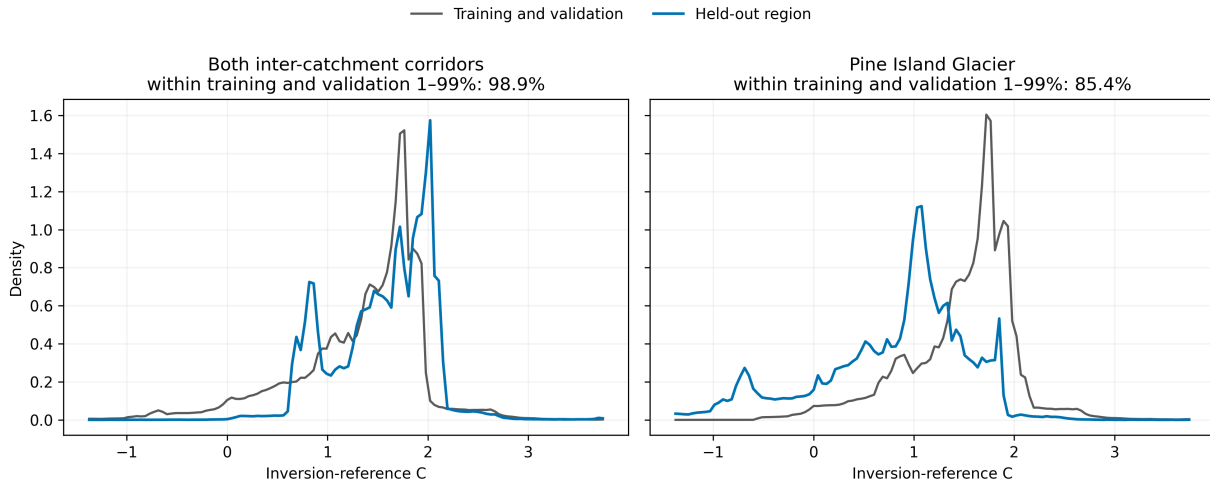


Fig. 11. Reference- C distributions for the two regional tests. Each panel compares the training and validation rows with the complete held-out region. The percentage shown is the fraction of held-out C values that fall within the 1st–99th-percentile range of the training and validation data.

Table 5. Reference- C distributions in the training-and-validation and held-out regions. Intervals are the 1st–99th percentiles of dimensionless C . The final column gives the percentage of held-out rows whose C values fall within the corresponding training-and-validation interval.

Experiment	Training and validation rows	Held-out rows	Training and validation interval	Held-out interval	Held-out within training range (%)
SQ01	1,448,295	12,321	−0.22–2.67	−0.76–1.30	51.9
SQ02	1,447,471	12,321	−0.69–2.67	1.43–1.96	100.0
SQ03	1,450,403	12,432	−0.69–2.67	0.21–1.40	100.0
SQ04	1,447,471	12,321	−0.69–2.67	1.80–1.96	100.0
SQ05	1,459,429	12,321	−0.65–2.66	0.84–2.35	100.0
SQ06	1,447,471	12,321	−0.69–2.67	0.28–1.43	100.0
SQ07	1,447,471	12,321	−0.69–2.67	0.95–1.55	100.0
SQ08	1,453,825	12,321	−0.69–2.66	1.18–1.71	100.0
SQ09	1,458,393	12,321	−0.69–2.66	0.62–1.23	100.0
SQ10	1,463,861	12,321	−0.69–2.66	1.50–1.81	100.0
Inter-catchment	1,241,683	289,309	−0.72–2.65	0.49–2.66	98.9
PIG	1,303,307	227,685	−0.08–2.69	−1.28–2.05	85.4

agreement on randomly withheld rows within the training region does not imply comparable transfer to a spatially separated square.

We also compare the predicted C fields directly with the reference C to measure how closely the MLP reproduces the inversion result. Table 7 reports C RMSE and R_C^2 for every central square. We retain negative R_C^2 values; a negative value means that the predicted C has larger squared error than using the mean reference C within that square. The median R_C^2 values across the ten squares are −1.39, −1.35, −7.97, −6.52, −1.41, and −1.21 for CFG01–CFG06, respectively. Poor agreement with the reference C can still coexist with improved modeled velocity, so these statistics answer a different question from the velocity comparison.

For the PIG holdout experiment, the median validation C RMSE / R_C^2 across the ten MLP members is 0.081/0.981 for CFG01, 0.077/0.983 for CFG02, and 0.056/0.991 for CFG03. These values were calculated on each member’s validation rows outside the withheld PIG region. Despite this strong validation agreement, the corresponding median predicted fields have PIG holdout C RMSE / R_C^2 values of 0.92/−0.49, 0.85/−0.28, and 1.23/−1.69, respectively. Thus, performance on randomly withheld validation rows does not predict spatial transfer to the withheld catchment.

The inter-catchment experiment provides an additional regional test by withholding the complete PIG–

Table 6. Validation agreement with the inversion-reference control. Each entry is the median C RMSE / R_C^2 across the ten MLP members for that square-configuration experiment, calculated on each member’s own validation rows. RMSE is in the dimensionless control C , and R_C^2 is dimensionless. The validation rows lie outside the excluded 130 km footprint and were used to select the fitted MLP; they are not independent test data.

Square	CFG01	CFG02	CFG03	CFG04	CFG05	CFG06
SQ01	0.077/0.984	0.075/0.985	0.056/0.992	0.363/0.652	0.020/0.999	0.022/0.999
SQ02	0.078/0.987	0.081/0.986	0.057/0.993	0.387/0.680	0.023/0.999	0.022/0.999
SQ03	0.072/0.988	0.077/0.986	0.070/0.989	0.383/0.661	0.022/0.999	0.021/0.999
SQ04	0.080/0.986	0.079/0.986	0.056/0.993	0.393/0.666	0.023/0.999	0.022/0.999
SQ05	0.076/0.987	0.077/0.986	0.055/0.993	0.364/0.697	0.023/0.999	0.021/0.999
SQ06	0.073/0.988	0.080/0.986	0.057/0.993	0.382/0.681	0.022/0.999	0.022/0.999
SQ07	0.077/0.987	0.077/0.987	0.058/0.993	0.388/0.681	0.021/0.999	0.022/0.999
SQ08	0.077/0.988	0.079/0.987	0.056/0.993	0.397/0.669	0.022/0.999	0.023/0.999
SQ09	0.077/0.987	0.079/0.986	0.057/0.993	0.384/0.676	0.022/0.999	0.022/0.999
SQ10	0.077/0.987	0.080/0.986	0.058/0.993	0.391/0.676	0.023/0.999	0.022/0.999

Table 7. Comparison between predicted and reference C in the ten central squares. Each entry gives C RMSE / R_C^2 for the median predicted C field, computed on finite-element cells and weighted by cell area. RMSE is in dimensionless C units. Values quoted on the 450 m observation rows elsewhere in the text differ by at most 0.01.

Square	CFG01	CFG02	CFG03	CFG04	CFG05	CFG06
SQ01	0.42/0.25	0.41/0.29	1.44/−8.00	1.21/−5.42	0.58/−0.47	0.52/−0.20
SQ02	0.22/−2.69	0.22/−2.46	0.88/−56.51	0.61/−26.00	0.17/−1.04	0.15/−0.69
SQ03	0.38/−0.63	0.37/−0.61	0.54/−2.31	0.54/−2.32	0.35/−0.38	0.33/−0.29
SQ04	0.20/−29.81	0.20/−30.02	0.36/−98.61	0.33/−80.67	0.16/−18.76	0.16/−18.71
SQ05	0.37/0.03	0.37/0.02	0.46/−0.50	0.48/−0.63	0.41/−0.21	0.42/−0.25
SQ06	0.52/−1.83	0.51/−1.72	0.51/−1.77	0.47/−1.35	0.33/−0.14	0.34/−0.19
SQ07	0.24/−2.44	0.23/−2.18	0.57/−17.72	0.52/−14.99	0.30/−4.13	0.31/−4.77
SQ08	0.20/−0.96	0.20/−0.98	0.28/−2.78	0.31/−3.43	0.24/−1.77	0.24/−1.73
SQ09	0.74/−18.37	0.74/−18.49	0.87/−26.27	0.49/−7.63	0.53/−8.90	0.51/−8.38
SQ10	0.10/−0.79	0.10/−0.83	0.23/−7.93	0.31/−14.95	0.31/−15.21	0.27/−11.59

Thwaites and Thwaites–Dotson corridors. This tests transfer from the named glacier catchments to the regions between them. We trained CFG04–CFG06 for this experiment, so it is not directly comparable with the six-configuration comparison in the ten central squares. For the two corridors together, the holdout C RMSE / R_C^2 values are 0.81/−1.86, 0.33/0.53, and 0.33/0.52 for CFG04–CFG06. Across both corridors, the median predicted C fields give velocity RMSE values of 470.3, 119.0, and 109.3 m a^{−1} for CFG04, CFG05, and CFG06, respectively, compared with 578.4 m a^{−1} for uniform C and 28.1 m a^{−1} for the inversion reference. The corresponding relative RMSE values are 0.81, 0.21, and 0.19. The two corridors are spatially separate, and the calculation excludes the intervening region. All three configurations reduce velocity RMSE relative to uniform C over the combined corridor area, with lower errors for CFG05 and CFG06 than for CFG04. The ten central squares show the same ordering: CFG05 and CFG06 each have lower velocity RMSE than CFG04 in seven of the ten squares and lower median relative RMSE (0.59 and 0.61 against 0.88; Table 2), although the margin is much larger over the corridors. For these velocity calculations, all four controls replace the inversion reference only inside the combined corridor geography and retain the same reference C outside it, matching the controlled comparison used for the square and PIG CFG02 results. These are the most favourable holdout conditions tested here: 98.9% of the held-out corridor C values fall within the training-and-validation 1st–99th-percentile interval, compared with 85.4% for PIG (Table 5), and the withheld corridors are bounded on both sides by catchments that remain in the training and validation dataset. The corridor values therefore indicate what the point-wise mapping achieves under favourable conditions rather than general transferability, and even there the best predicted controls, CFG05 and CFG06, leave velocity RMSE about four times that of the inversion reference.

Finally, we examine the small number of unusually large reference- C values because mean-square loss weights large prediction residuals more heavily. Across all eligible rows, C has a median of 1.54, a 1st–99th-percentile range of −0.68 to 2.65, a 99.9th percentile of 3.76, and a maximum of 24.18. Values greater than 5 occur in only 821 of 1,530,992 rows (0.054%). They are more common near the grounding transition but are not confined to it: 261 occur where $0.1 < \phi \leq 0.2$. The scaling used for training changes the numerical scale of C but does not selectively reduce the contribution of large prediction residuals. The largest 0.1% of squared deviations from the target median account for 12.5% of the total squared deviation.

For the MLPs selected by minimum validation MSE, rows with $C > 5$ account for a median of 1.59% of validation squared error; excluding them from validation scoring reduces median validation RMSE by 0.77% and does not materially alter the configuration comparisons. No such rows occur in the ten central

squares or the PIG evaluation region, so we retain them rather than introduce a target-based exclusion and retraining step. These rare large values are features of the inversion-derived target and should not be interpreted as direct evidence for extreme physical basal resistance.

Training representation and prediction error

We also test whether prediction errors are related to how densely the predictor values at each held-out location are represented in the training data. For CFG02 and CFG04 in the ten central 50 km squares, and CFG02 in PIG, we calculate d_{20} for each held-out row, defined as the distance to its 20th-nearest training row. Nearest-neighbor distances underlie multivariate density estimation (Loftsgaarden and Quesenberry, 1965); here d_{20} describes training representation rather than estimating a probability density. Distances are calculated after applying the same transformation used for joint support: predictor ranks are mapped to normal scores, followed by PCA retaining at least 99% of the variance, with the retained components scaled to unit variance. We fit this transformation to the eligible sector-wide 5 km grid. Smaller d_{20} values indicate that many similar predictor combinations occur in the training data, whereas larger values indicate sparser representation. Unlike the joint-support calculation, we impose no minimum geographic separation here; we use all training rows, including nearby rows and repeated predictor combinations.

For each MLP realization, we sample 10,000 training rows uniformly without replacement. We calculate the d_{20} value for each sampled training row against the complete training set, excluding only that row itself. These 10,000 values provide a reference distribution for the held-out rows. Each held-out d_{20} value is assigned the percentile

$$p = 100 \frac{N(d_{20,\text{train}} < d_{20,\text{test}}) + \frac{1}{2}N(d_{20,\text{train}} = d_{20,\text{test}})}{10,000}. \quad (12)$$

For each held-out row, we use the median percentile across the ten MLP realizations for that experiment and configuration. Higher percentiles indicate sparser representation in the training data. We group the held-out rows into three categories: $p \leq 50$, $50 < p \leq 95$, and $p > 95$. We fixed the neighbor count and category boundaries before examining prediction errors.

We examine the original training data without rebalancing. The predictors are continuous and several are physically or mathematically related. Balancing each predictor separately would not necessarily balance combinations of predictors and could change the relative frequency of the combinations present in the original data. Instead, this analysis measures how densely the existing training data represent each held-

out predictor combination.

To compare training representation with prediction error, we evaluate the median predicted C field on the same 450 m observation-grid rows used for the representation analysis. We first calculate the median on the model mesh, then interpolate it to the observation grid as we do for the reference C . All central-square rows are retained. The PIG analysis uses the same procedure over its complete eligible held-out area. Within each representation category, we calculate C RMSE, mean signed difference, and mean absolute difference from the reference C . Velocity RMSE and relative RMSE are calculated over the same rows.

The whole-square C RMSE values calculated on the regular 450 m grid differ by at most 2.37% from the area-weighted values in Table 7, which are calculated on the finite-element mesh. All C errors are reported in the original dimensionless C units.

Table 8 tests whether C prediction error increases as training representation becomes sparser. For CFG02, C RMSE increases from the better-represented to the more sparsely represented categories in seven of the ten squares. However, this pattern is not consistent across all experiments. The median C RMSE across the ten squares is 0.299, 0.297, and 0.301 for $p \leq 50$, $50 < p \leq 95$, and $p > 95$, respectively, and the median within-square Spearman correlation between representation percentile and absolute C error is only 0.069. Thus, although some individual squares show larger C errors where similar training examples are sparser, training representation does not generally explain C prediction error. CFG04 shows even less evidence of such a relationship: only one square has increasing C RMSE across the three categories, and the median within-square correlation is -0.052 .

To test whether differences in ice speed obscure the weak relationship between training representation and C error, we repeat the analysis within observed-speed classes. For each class, we compute the correlation between representation percentile and absolute C error. Below 100 m a^{-1} , the median within-square correlations are 0.080 for CFG02 and -0.113 for CFG04. Between 100 and 500 m a^{-1} , they are 0.079 and 0.015, respectively. These values remain close to zero, indicating that sparser training representation is not consistently associated with larger C error within either the slow- or moderate-flow classes. Only one square contributes to the $500\text{--}1000 \text{ m a}^{-1}$ class, so there are not enough repeated spatial tests to draw the same conclusion for faster flow.

Relative velocity RMSE also does not vary consistently with training representation. In SQ06, CFG02 performs worse than uniform C in the best-represented category (relative velocity RMSE 1.15) but improves on uniform C in the sparsest category (0.96). In SQ05, C RMSE decreases from 0.416 to 0.337 between

Table 8. Prediction error in C relative to the inversion reference and training representation in the central squares. The three columns give dimensionless C RMSE for increasingly sparse representation categories. Spearman ρ relates the representation percentile of each row to its absolute C error within each square. The last row for each configuration gives the median across the ten squares.

Configuration	Square	$p \leq 50$	$50 < p \leq 95$	$p > 95$	ρ
CFG02	SQ01	0.371	0.410	0.418	0.076
	SQ02	0.206	0.232	0.265	0.107
	SQ03	0.370	0.384	0.411	0.086
	SQ04	0.201	0.204	0.211	0.056
	SQ05	0.416	0.351	0.337	-0.141
	SQ06	0.493	0.516	0.543	0.035
	SQ07	0.227	0.242	0.245	0.090
	SQ08	0.175	0.212	0.198	0.133
	SQ09	0.731	0.735	0.753	0.062
	SQ10	0.130	0.103	0.110	-0.087
	Median	0.299	0.297	0.301	0.069
CFG04	SQ01	1.100	1.298	1.058	-0.101
	SQ02	0.608	0.634	0.470	-0.049
	SQ03	0.512	0.572	0.587	0.139
	SQ04	0.375	0.364	0.236	-0.637
	SQ05	0.473	0.508	0.422	0.065
	SQ06	0.383	0.515	0.474	0.224
	SQ07	0.634	0.477	0.466	-0.257
	SQ08	0.348	0.307	0.309	-0.044
	SQ09	0.517	0.491	0.411	-0.055
	SQ10	0.299	0.317	0.302	-0.094
	Median	0.492	0.499	0.444	-0.052

these categories (Table 8), while relative velocity RMSE increases from 0.37 to 0.81. In SQ02–CFG04, the best-represented category has a C RMSE of 0.608 and a relative velocity RMSE of 1.45, whereas the sparsest category has lower C RMSE (0.470) and relative velocity RMSE (1.34). These examples show that sparser training representation does not consistently correspond to larger relative velocity RMSE. They also show that closer agreement with the reference C does not necessarily produce lower relative velocity RMSE.

MLP TRAINING AND STATISTICAL COMPARISONS

This section describes the MLP architecture and training procedure, the repeated training realizations used for each experiment, and the statistical comparisons used to summarize differences among configurations.

The MLP has ten hidden layers of 200 units. Each hidden layer is followed by batch normalization and a SiLU activation, and the final layer is linear. Predictors and C are robustly scaled using medians and interquartile ranges fitted on training rows only. Adam uses an initial learning rate of 10^{-3} and batch size 1024, with reproducible epoch-wise shuffling. The learning rate is multiplied by 0.2, down to 10^{-6} , after 15 epochs without a validation-MSE decrease of at least 10^{-4} . Training stops after 45 epochs without such a decrease, and the fitted MLP with the lowest unpenalized validation MSE is used.

A separate CFG06 calculation selected the L2 penalty without using any of the ten central squares. Its fixed 70/20/10 split contained 1,407,671 rows: 985,369 for training, 281,534 for validation, and 140,768 for testing. We applied L2 penalties of 0, 10^{-6} , 10^{-5} , and 10^{-4} to the hidden-layer weights and compared them using unpenalized validation MSE. The respective minimum values were 0.001207, 0.001332, 0.001583, and 0.003117. No nonzero penalty reduced validation MSE below that of the zero-penalty model, so all final MLP trainings used $\lambda_{L2} = 0$. We did not use the test rows to choose the penalty.

We did not optimize the network architecture separately for each predictor configuration or held-out region. Instead, we used the same relatively large network, with ten hidden layers of 200 units, for all experiments. This architecture provides substantial flexibility to represent nonlinear relationships between the predictors and C , while keeping the network design fixed across configurations and spatial tests. We limit overfitting through a separate validation set, learning-rate reduction, early stopping, and restoring the model with the lowest validation MSE. The training and validation histories shown below provide an additional check. A full architecture search would require separate tuning across multiple predictor configurations and spatial splits and was beyond this study’s scope. The chosen architecture

should therefore be viewed as a fixed, flexible modeling choice rather than an optimized architecture.

Each experiment–configuration pair uses ten reproducible training realizations with different training/validation row assignments, initial weights, and epoch shuffles. We trained 660 MLPs in total. The regional tests use CFG01–CFG03 for PIG and CFG04–CFG06 for the inter-catchment corridors. For each experiment–configuration pair, we combine the ten predicted C fields by taking the median at each control point, yielding 66 median fields. The controlled forward evaluation in the main text uses the 60 square fields and the PIG CFG02 field. We use the individual realizations to assess sensitivity to the training procedure and to construct the median prediction; they are not treated as independent spatial tests or as estimates of predictive uncertainty.

For the velocity-error calculations, we interpolate modeled velocity to the pixel centers of the regular 450 m MEaSUREs observation grid and compare it with the original paired raster components there. We average squared vector residuals over the retained grid points before taking the square root. Because the observation grid has uniform spacing, this provides an equal-area approximation to the spatial mean rather than an exact finite-element integration.

For the spatially uniform- C reference, we calculate the mean inversion-reference C from finite-element cells in the non-held-out region. We weight cell means of C by cell area. We evaluate geographic membership and the criterion $\phi > 0.1$ at cell centers, also requiring positive thickness and finite C . The resulting value therefore uses model cells rather than observation-grid row means. It is applied over the same withheld footprint or catchment used for the predicted C field, with inversion-reference C retained elsewhere.

The ten squares are treated as the independent spatial units for statistical comparison. The PIG and inter-catchment experiments are secondary regional tests and do not add independent units to these comparisons. The spatially uniform- C simulation tests whether predictor-dependent spatial variation reduces velocity RMSE relative to uniform C ; it is not a comparison with another learning algorithm. We report results for every square and summarize them using the median and interquartile range across the ten squares. Exact paired sign tests compare predictor configurations chosen in advance. We do not count the individual MLP realizations, or the central square and full footprint from the same experiment, as additional independent units. All 66 median predicted C fields are finite. We did not clip predictions to the range of the inversion-reference C .

The complete histories in Fig. 12 show that the training and validation losses decrease and then approach a plateau. Median validation MSE decreases from its initial value by factors of 19.5, 19.4, 37.7, 1.68, 148,

and 162 for CFG01–CFG06, respectively. After the early epochs, median validation MSE stays below median training MSE, ending at about half of it for CFG03, CFG05, and CFG06. Training MSE is higher because it is computed during each epoch, with batch normalization using the statistics of each batch rather than the running averages used for validation. The histories therefore provide no clear evidence that the relatively large network is increasingly fitting the training rows at the expense of the validation rows. Validation agreement with the reference C is high for CFG01, CFG02, CFG03, CFG05, and CFG06, while CFG04 shows a much smaller reduction in validation loss and higher validation C RMSE and lower R_C^2 . However, CFG04 still reduces velocity RMSE relative to uniform C in seven of the ten spatially withheld squares. The main implication is that fitting C well on random validation rows within the training region does not, by itself, predict performance in a spatially separated region. The common architecture and training procedure provide a controlled comparison among predictor configurations, while the spatially withheld tests assess whether the learned relationships produce useful modeled velocities elsewhere.

We used paired sign tests to assess whether one configuration consistently had lower velocity RMSE than another across the ten withheld squares (Hollander and others, 2013, Chapter 3, Section 3.4). We tested all 15 possible pairwise configuration comparisons; none was statistically significant after Holm correction for multiple comparisons (Holm, 1979). All 15 Holm-adjusted p -values are 1.0. For the five comparisons corresponding directly to predictor additions or removals in the configuration design, the unadjusted two-sided p -values were 1.0 for CFG02–CFG01, 1.0 for CFG04–CFG03, 0.344 for CFG05–CFG02, 0.344 for CFG05–CFG04, and 1.0 for CFG06–CFG05. Thus, although some configurations have lower median RMSE or perform better in more squares, no configuration shows a statistically significant consistent advantage over another across the ten spatial tests. With ten squares and 15 comparisons, a difference reaches significance only if one configuration has lower RMSE in all ten squares. The tests therefore show that no configuration is better everywhere, but they cannot distinguish smaller differences, such as one configuration being better in seven or eight squares, from chance.

BedMachine reconstruction and velocity error

Because several predictors are derived from BedMachine, we also examined whether differences in modeled velocity performance could be related to the BedMachine reconstruction itself. We considered two quantities: the source class, which identifies the method used to construct the bed elevation, and the reported bed-elevation error. We did not use either quantity as an MLP predictor or to select or exclude evaluation

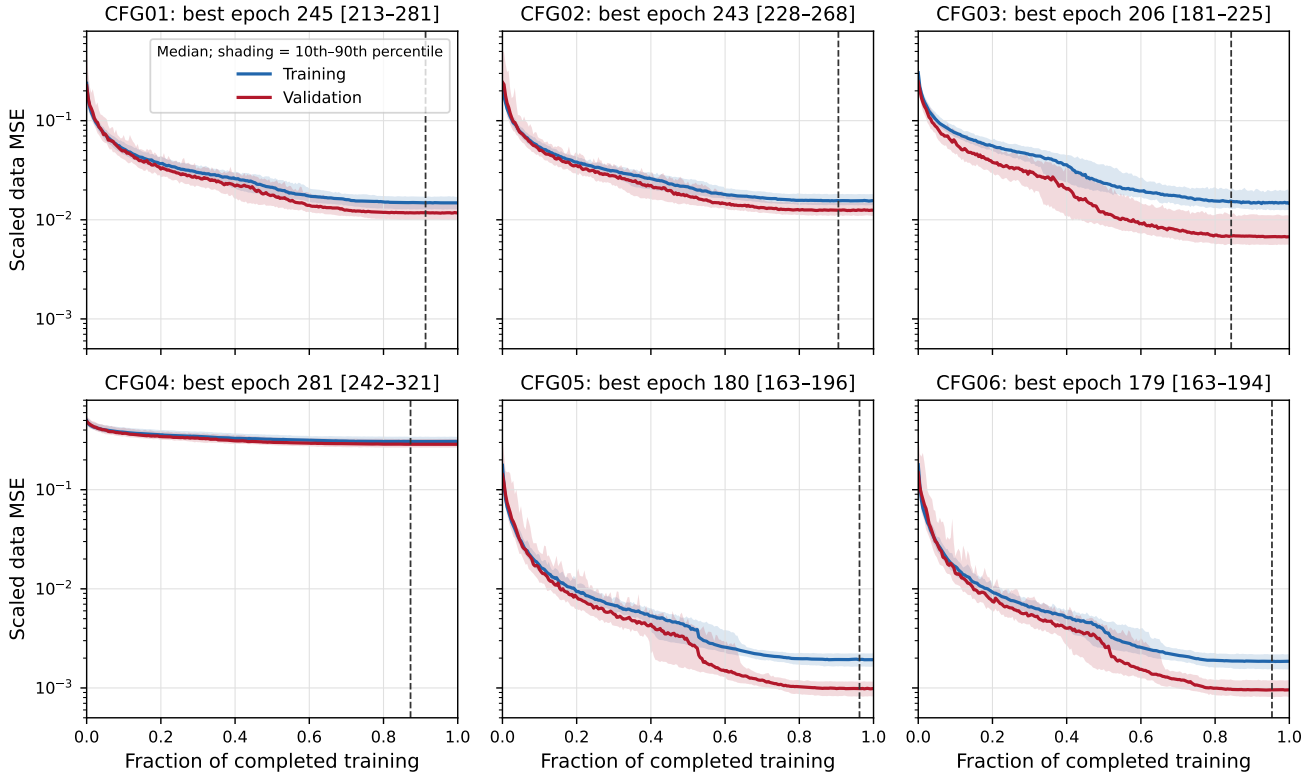


Fig. 12. Training histories for all 600 primary-square MLP runs, comprising ten squares and ten MLPs for each configuration. To retain every run in the summary, we scale each history from its first epoch (training fraction zero) to the end of training (training fraction one). Lines show median training and validation data MSE, and shading spans the 10th–90th percentiles across the 100 runs in each configuration. Training MSE is computed during each epoch with batch normalization in its training mode and lies above validation MSE after the early epochs. The vertical dashed line marks the median position of the fitted MLP with the lowest validation MSE; panel titles give the median and interquartile range of its absolute epoch. MSE is computed after robust scaling fitted only on the training rows and is used to fit the MLP, not to measure held-out modeled velocity performance. For each experiment, we did not use its central square to select a fitted MLP.

Table 9. Spearman correlations between reported BedMachine bed-elevation error and local velocity error in the ten central squares. Each configuration is summarized by the median and range of its ten within-square correlations.

Configuration	Median	Minimum	Maximum
CFG01	-0.005	-0.117	0.139
CFG02	-0.018	-0.135	0.137
CFG03	0.012	-0.151	0.082
CFG04	0.041	-0.139	0.138
CFG05	-0.062	-0.218	0.193
CFG06	-0.059	-0.214	0.181

rows. The analysis uses the ten central 50 km squares and the velocity from each configuration’s median- C forward simulation.

For each square and configuration, we calculated the Spearman correlation between reported bed-elevation error and local velocity-vector error, $\|\mathbf{u}_{\text{ML}} - \mathbf{u}_{\text{obs}}\|$. Table 9 summarizes the ten within-square correlations. Their small magnitudes and changing signs show that larger reported bed-elevation errors did not consistently coincide with larger velocity errors.

BedMachine source class records how the geometry supplied to the MLP was reconstructed. Class 5 denotes streamline-diffusion interpolation, which smooths thickness preferentially along flow; it is not a direct bed measurement. Eight central squares contain only class 5 rows. SQ05 contains 838 mass-conservation rows (class 2), 174 interpolation rows (class 3), and 11,309 class 5 rows; SQ06 contains 12, 18, and 12,291 rows in these classes, respectively. These are counts of evaluated observation-grid rows.

Because reconstruction method changes little within most squares, these tests cannot separate its effect from other differences between locations. Performance also varies among squares using the same method, so differences between source classes cannot alone account for the observed spatial variation. We did not test source class as an additional predictor or quantify how much along-flow smoothing changes the predictor gradients and modeled velocity.

Predicting where transfer succeeds

The preceding diagnostics examine predictor support, training representation, agreement with the reference C , and the BedMachine reconstruction one at a time. Here we test whether the predictors, taken together, can indicate in advance where the predicted C improves modeled velocity. This is the information a user would need to decide where a point-wise prediction can be relied upon.

For each observation-grid row in a withheld region, we recorded whether the predicted C gives lower local velocity error than uniform C , $e_{\text{ML}} < e_{\text{uniform}}$, using the local vector-error magnitude defined in the Methods. We also used a stricter criterion, $e_{\text{ML}} < 0.5 e_{\text{uniform}}$, which requires the predicted C to at least halve the local error. We then trained a random forest classifier to predict each outcome. A random forest combines many decision trees fitted to random subsets of the data and can represent nonlinear relationships among many inputs without specifying their form (Breiman, 2001). We used three input sets: observed speed alone, the twelve predictors, and the twelve predictors together with observed speed. We included observed speed because any user can obtain it from the velocity map. If the predictors contain additional information about where transfer succeeds, the combined classifier should outperform the classifier that uses observed speed alone. We used the same classifier settings for every test: 120 trees, a maximum depth of 14, at least 40 rows in each leaf, and the two outcomes given equal total weight regardless of how often each occurs. Equal weighting makes the classifier predict the less common outcome more often than its frequency alone would suggest; the cut-off tests below show that this does not explain the results.

We evaluated each classifier on a region it had not seen. For the squares, we trained on the central squares of nine tests and evaluated on the central square of the tenth, repeating this for every square and configuration. Every training row therefore comes from a predicted C field produced without that location, as in the evaluated square. For PIG, we trained on the ten central squares, excluding 18,409 rows that lie inside PIG, and evaluated CFG02 over the complete catchment. We report the percentage of held-out rows the classifier labels correctly, and compare it with the percentage obtained by predicting whichever outcome is more common in that region. That comparison is the relevant one: a user who knew nothing beyond the regional outcome rate could achieve it without any predictors.

In 10 of the 60 square tests (12 for the stricter criterion), the outcome was the same at every evaluated row, so no classifier could separate locations within the square. In SQ04, for example, the predicted C improved on uniform C at every row for all six configurations. We report these tests separately below; in them, success or failure is a property of the whole square rather than of individual locations.

Across the 294 remaining tests the classifiers labeled 1,967,017 of 3,626,370 held-out rows correctly, or 54.2%. The median was 51.4%, against 77.5% for predicting the more common outcome, and the classifiers exceeded that rule in 32 of the 294 tests. For the improvement criterion the median percentages correct were 57.9 using observed speed alone, 62.6 using the twelve predictors, and 61.6 using both, against a median of 77.6 for the more common outcome (Fig. 13a). For the stricter criterion they were 59.2, 39.6 and

39.1, against 74.3 (Fig. 13b). Adding the twelve predictors to observed speed did not help: the combined classifier exceeded the more common outcome in 5 of 50 tests under the improvement criterion and 4 of 48 under the stricter one, while observed speed alone did so in 10 and 8.

These failures are not a matter of the classifier fitting its training data poorly. The same classifier labeled 98.7% of its training rows correctly, and 98.3% of rows held out at random from within the training squares, compared with about half of the rows in a spatially separate square. Nor do they depend on where the classifier's output is cut into a yes or no decision. Choosing that cut-off on rows held out from the training squares, as a user could, raised the median to 54.1% and exceeded the more common outcome in 29 of the 294 tests. Even choosing the best possible cut-off for each withheld square in hindsight, which no user could do, raised the median to 81.6% and exceeded the more common outcome in only 122 of the 294 tests.

The classifiers also failed to transfer to PIG, where 227,685 rows were evaluated. For the improvement criterion they labeled 30.8%, 36.4% and 29.9% of rows correctly for the three input sets, against 77.6% for the more common outcome. For the stricter criterion they labeled 64.2%, 90.8% and 89.7% correctly, against 92.0%. No input set exceeded the more common outcome in either case.

Part of the reason is that the relationship between ice speed and success differs between the two settings. In the rows used to train the PIG classifier, the predicted C meets the improvement criterion at 98.5% of rows in the slowest quarter of speeds but at only 48.3% and 52.1% in the two fastest quarters. In PIG the pattern is reversed: the criterion is met at 51.3% of rows in the slowest quarter of PIG speeds and 93.8% in the fastest. No training row for the PIG test is faster than 455 m a^{-1} , 11% of the evaluated PIG area is faster than this, and the improvement criterion is met at 76.2% of the PIG rows within the training speed range against 88.8% of those beyond it. A classifier that has learned where transfer succeeds in the squares therefore carries that association in the wrong direction when it is applied to the catchment.

The predictors therefore do not provide a reliable way to identify in advance where the predicted C will improve modeled velocity. The combined classifier is not consistently better than one using observed speed alone, rarely improves on predicting the more common regional outcome, and does not transfer from the squares to PIG. That the same classifier reproduces its training squares almost exactly points to the difficulty being spatial transfer rather than model capacity, and the tests in which every row shared one outcome point the same way: whether the point-wise relationship succeeds is largely a property of the withheld region, of which there are ten here, rather than of individual locations within it. Together

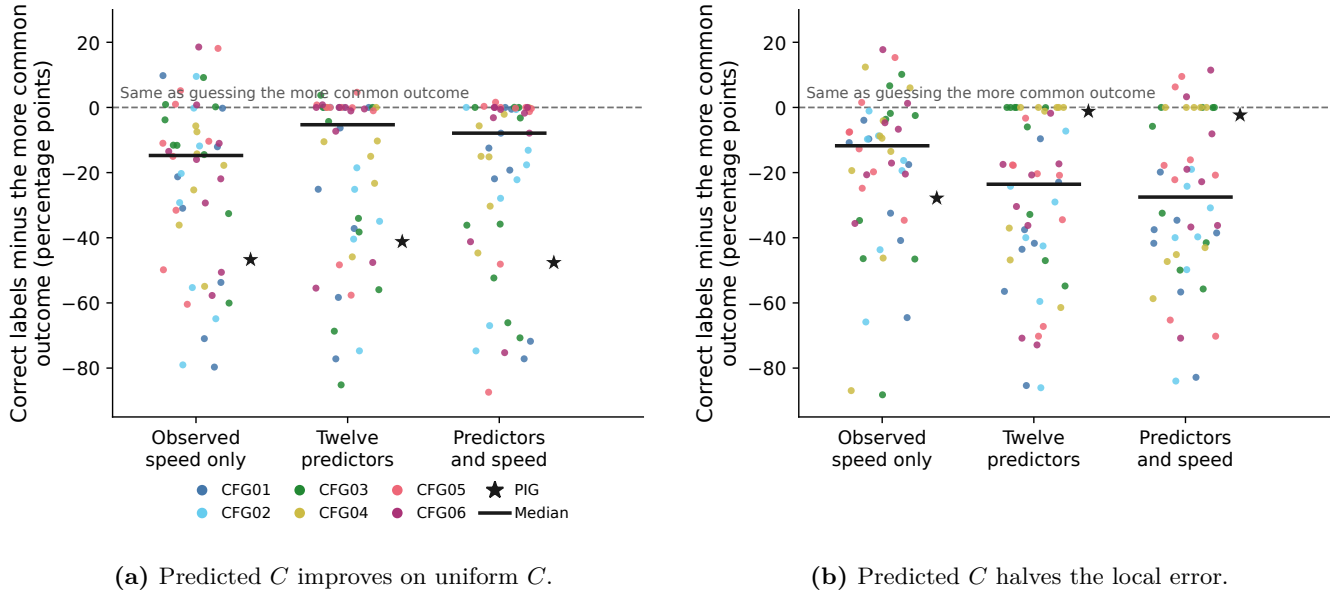


Fig. 13. Ability of a random forest classifier to predict where the predicted C improves on uniform C . Each point is one withheld square and configuration, and gives the percentage of held-out rows labeled correctly minus the percentage obtained by predicting whichever outcome is more common in that square; color identifies the configuration. Values below the dashed line are no better than that rule. The classifier is trained on the central squares of the other nine tests. Stars show the PIG CFG02 test, for which the classifier is trained on all ten central squares with rows inside PIG removed. Horizontal bars mark medians across the square tests. Panel (a) uses $e_{\text{ML}} < e_{\text{uniform}}$ and panel (b) uses $e_{\text{ML}} < 0.5 e_{\text{uniform}}$. We omit tests in which every row or no row met the criterion.

with the tests of predictor support and training representation, this indicates that the available predictors, individually or together, do not show where the point-wise relationship will transfer successfully.